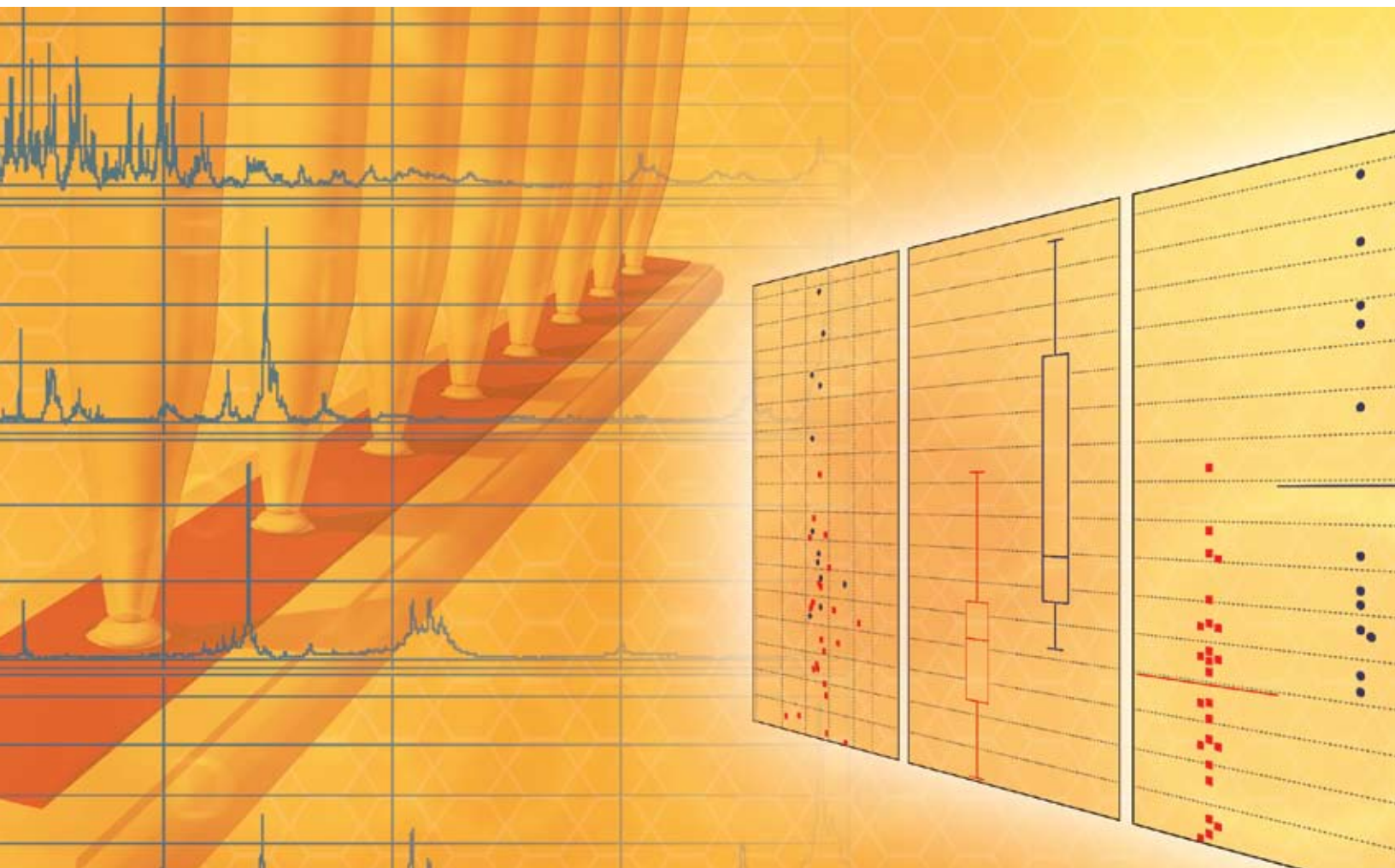




Biomarker Discovery Using SELDI Technology

A Guide to Data Processing and Analysis
Using ProteinChip® Data Manager Software

BIO-RAD

Data Processing and Analysis Using ProteinChip® Data Manager Software

About This Guide

This guide describes the concepts and workflow involved in the processing and analysis of data generated with the ProteinChip SELDI system. A content map and summary of the general workflow steps are provided at right. The workflow begins with the organization and processing of the raw spectra that are obtained during data collection and ends with the multivariate statistical analysis methods available within ProteinChip data manager software.

The recommendations presented are based on the ProteinChip SELDI protein biomarker discovery process. They are intended as general workflows and settings for a majority of sample types and experiments. In some cases, modifications may be necessary. Included are recommendations for each stage of data processing and analysis, discussions of the different parameters and settings that may be used, and examples of how each step may affect final data analysis.

A more detailed summary of the workflow steps is provided in the Appendix. Details for performing each step in ProteinChip data manager software, however, are not provided; for this information, refer to the ProteinChip Data Manager Software Operation Manual.

Data Organization

Data Organization

- Ensure proper annotation of spectra
- Group spectra into folders

Mass Calibration and Data Processing

Mass Calibration

- Create the calibration equation using calibration standards
- Improve mass accuracy through external calibration

Processing Spectral Data

- Adjust peak intensity calculations using baseline subtraction
- Reduce noise by filter adjustment
- Improve the signal-to-noise ratio by setting the noise range

Processing Conditions

- Standardize intensities by normalization
- Improve mass precision by spectrum alignment

Generation and Analysis of Peak Clusters

Generation of Peak Clusters

- Associate peaks from multiple spectra into peak clusters
- Adjust peaks within clusters using cluster editing

Selection of Candidate Biomarkers

- Select biomarker candidates using P values
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Assessment of Sample Relationships

- Examine sample hierarchical clustering patterns
- Visualize sample similarity using PCA

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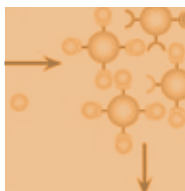
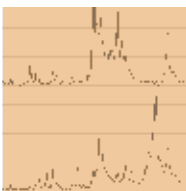
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Part I: The Data Analysis Process

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The Data Analysis Process

Mass spectrometry (MS)-based clinical proteomics and biomarker research generate vast and complex data sets. As a result, data analysis can become a critical bottleneck in the research process. This chapter provides an overview of the data analysis workflow used with the ProteinChip SELDI system and introduces the primary types of ProteinChip SELDI data.

Overview

The MS-based techniques used in biomarker research generate vast quantities of data. For example, depending on the type of study and sample source, discovery phase experiments may require 100 or more samples and profile thousands of proteins and peptides, all with varying expression levels. Rigorous data analysis is required to sort through these data to select the most robust candidate biomarkers.

The data analysis workflow used for biomarker discovery with the ProteinChip SELDI system and ProteinChip data manager software involves the following steps:

- Data organization — creation or revision of sample annotations and organization of spectra into folders. Analysis of one condition requires organizing spectra from that condition into the same folder
- Mass calibration — generation or updating of the calibration equation, which converts raw time-of-flight (TOF) data into mass data
- Processing — baseline subtraction, filtering, noise estimation, alignment, and normalization of spectra. These steps optimize the accuracy and reproducibility of peak mass and intensity measurements prior to clustering

- Generation of peak clusters — creation of peak clusters from spectral data, cluster editing, and preparation of peak cluster data for statistical analysis. Individual spectral features or peaks are labeled across all spectra in a folder and then grouped together based on matches between their calculated masses. The result is a list of masses, or peak clusters, that contain the same peaks from each spectrum with associated intensities for each sample
- Selection of candidate biomarkers — analysis of peak clusters by univariate statistical methods to identify candidate biomarkers that distinguish sample groups. The comparison of peak intensities of the same peak across multiple spectra (within a cluster) is a measure of protein expression levels from multiple samples
- Assessment of sample relationships — further evaluation of peak clusters using unsupervised multivariate statistics to reveal desired associations, examine possible sources of experimental variability (preanalytical or analytical bias), and generate more confidence in candidate biomarkers

This process yields lists of candidate biomarkers that, to effectively conclude the biomarker research process, must undergo experimental validation. In addition, biomarker candidates are often identified and more specific clinical assays are developed. For more information about the design and implementation of biomarker research projects, see the sidebar at the end of this chapter and refer to bulletin 5642, Biomarker Discovery Using SELDI Technology: A Guide to Successful Study and Experimental Design.

ProteinChip SELDI Data Types

The data analysis process is represented by the data types it generates.

SELDI Spectra

ProteinChip SELDI analysis begins with laser desorption and ionization of proteins from a ProteinChip array surface and detection by TOF-MS. Inside the ProteinChip SELDI reader, a nitrogen laser illuminates the sample, and the laser energy induces a change of state from the solid, crystalline phase into the gas phase (desorption) with an associated ionization of the protein molecules. Once in the gas phase, the protein ions accelerate away from the metal array upon application of a voltage differential. The voltage differential imparts the same energy to all of the analytes in the sample, resulting in mass-dependent flight times. Once the protein ions strike the detector, the detector records the TOF of the analyte (expressed as a mass-to-charge ratio, or m/z) and the intensity of the response, which is directly related to the amount of that specific analyte on the array surface.

ProteinChip data manager software displays the data as a spectrum, plotting the measured signal intensity on the y-axis and the calculated m/z or mass on the x-axis (Figure 1). (Since the charge on most ionized proteins and peptides is 1, the m/z calculated from the TOF of a particle is equivalent to its molecular mass plus 1 Da.) The SELDI spectrum is the graphical representation of a sample's protein profile.

Each spectrum has several notable features referred to in this guide. They are:

- Matrix attenuation range — region in which most of the signal is generated by the matrix. This signal is suppressed during data collection up to the designated matrix attenuation setting to increase detection sensitivity. This region is excluded from noise calculations and data analysis
- Low-mass range — region <20 kD. Peaks in this range are generally sharp and well-resolved. Data collection and analysis parameters for this range are optimized to maximize peak resolution
- High-mass range — region >20 kD. Peaks in this range are generally of lower intensity and are broader due to chemical heterogeneity. Data collection and analysis parameters for this range are optimized to maximize detection sensitivity

Peaks

Within each spectrum, peaks represent the protein or peptide components of a sample, and their intensities correspond to the amounts of the species in a sample (their expression levels).

Peaks are labeled either manually by the user or automatically by the software. In the manual method, the user applies the **Centroid** function of the software to click on and label peaks in the spectra. In the automatic method, the software identifies peaks by locating changes in intensity versus the noise that are greater than a specified detection threshold.

The software calculates peak intensity using the distance from the top of the peak to the calculated baseline (Figure 2). It calculates the mass by applying a calibration equation generated from a calibration spectrum (see Mass Calibration later in this guide).

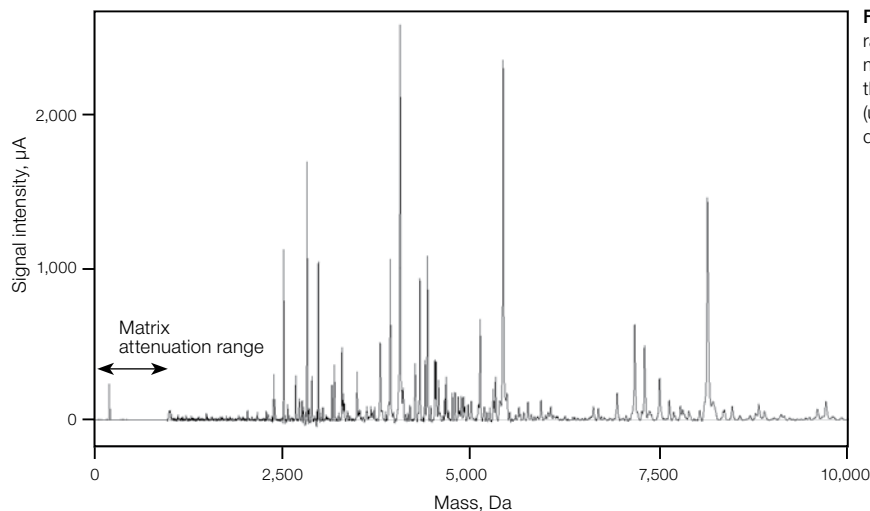


Fig. 1. SELDI spectrum. The 0–10 kD range of this spectrum is plotted with the mass on the x-axis and signal intensity on the y-axis. The matrix attenuation range (up to 1 kD in this example) is set during data acquisition.

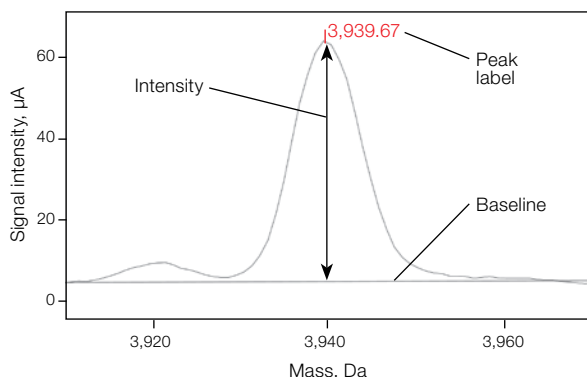


Fig. 2. Anatomy of a peak. Peaks are labeled either manually by the user or automatically by the software. The software calculates peak intensity using the distance from the top of the peak to the calculated baseline.

Peak Clusters

Though spectra and peaks may be two of the most recognizable forms of data generated by the ProteinChip SELDI system, peak clusters are the actual units of analysis used for biomarker discovery.

Peak clusters are groups of peaks of similar mass that are treated as the same protein or peptide across multiple spectra. ProteinChip data manager software defines clusters within the spectra in each data folder (Figure 3). Most of the data processing procedures described in this guide optimize the spectral data to enable detection of the most robust clusters and selection of the most robust biomarker candidates.

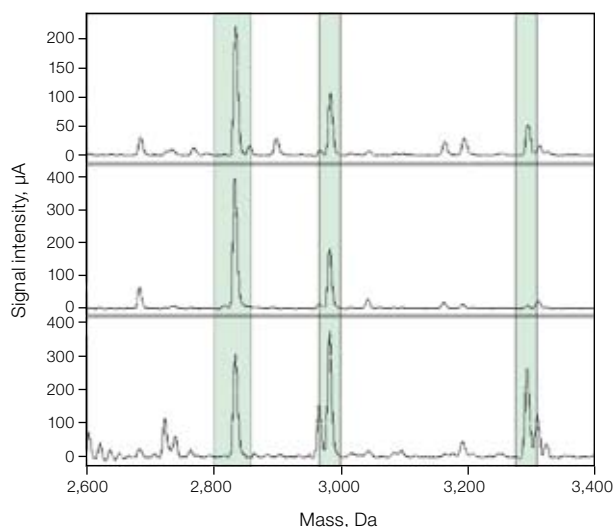


Fig. 3. Example of peak clusters. ProteinChip data manager software identifies peaks within the spectra in each data folder and groups them into clusters. In this example, three clusters are shown by the peaks that are grouped in each shaded box across three spectra.

For biomarker data analysis, the normalized peak intensities from each spectrum within a cluster are used to compare relative expression levels for that substance between different sample groups. The basis of biomarker analysis is searching for a difference in the expression level, or peak intensity, within a peak cluster (univariate analysis) or in a combination of multiple clusters (multivariate analysis).

Final Output

Univariate statistical analyses only consider single features. In biomarker research, this means they consider differences in the intensity of a single peak at a time. The aim is to identify peak clusters with statistically significant differences in intensity between sample groups (Figure 4). Evaluating clusters using univariate statistics (using the P values and areas under receiver-operating curves, or AUC values, calculated by ProteinChip data manager software) indicates their potential as single biomarkers.

The final output from ProteinChip data manager software is a list of candidate biomarkers for further analysis. The next steps in the characterization, confirmation, and application of the biomarker candidates involve experimental validation, identification, and clinical assay implementation as described in more detail in bulletin 5642, Biomarker Discovery Using SELDI Technology: A Guide to Successful Study and Experimental Design.

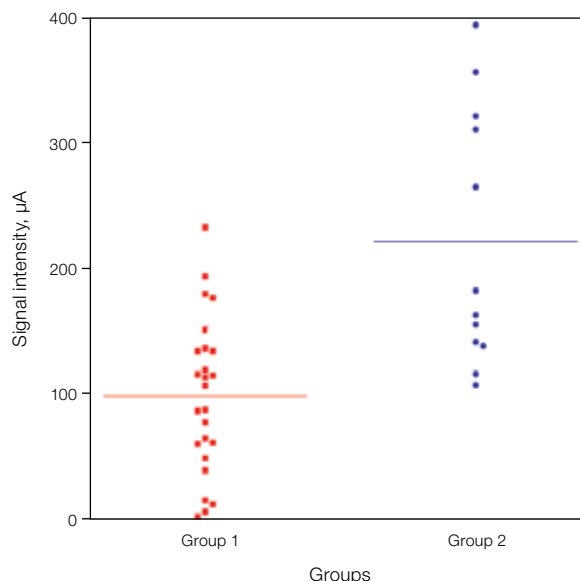


Fig. 4. Sample-group comparison within a cluster. Group scatter plot displaying the intensity values for one peak per spectrum from a selected cluster across two sample groups. The horizontal bars indicate the average intensity value for each group. In this example, the sample group at left contains the protein (peak) at expression levels that are statistically lower than those in the group at right. Such significant expression level changes may indicate the presence of a biomarker.

The Importance of Study and Experimental Design

Any analysis is only as good as the data it uses, and good spectral data require adherence to best practices in study and experimental design. To facilitate data



processing and analysis, and to maximize the potential for obtaining reliable biomarker candidates, follow the guidelines detailed in bulletin 5642, A Guide to Successful Study and Experimental Design. General recommendations are summarized below.

Minimize Preanalytical and Analytical Bias

These biases can have profound effects on the outcome of a study and on the ability to apply biomarker candidates to broader populations in validation studies or clinical assays. Sources of preanalytical bias include any systematic differences in patient populations or sample characteristics, including the procedures used for sample collection, handling, and storage. Sources of analytical bias arise from differences in how the samples are processed and analyzed.

Implement Standard Operating Procedures (SOPs)

Implement SOPs for all phases of an experiment, from sample collection to data analysis. Establishing and optimizing SOPs helps minimize bias and increases the opportunity to discover robust biomarkers.

Collaborate With Specialists

Collaborate with specialists, especially clinicians and biostatisticians, during the study design and data analysis phases of a project. All MS methods, including SELDI, generate vast and complex data sets that must be processed and analyzed properly to generate reproducible results of clinical utility. These profiling techniques generate many peak intensity features per sample, significantly more than the total number of samples in a study. Therefore, employ assistance from a biostatistician during the planning stages of a project and throughout the later phases of data analysis and evaluation of biomarker candidates.

Use the biostatistician's expertise to:

- Calculate the number of samples required for statistical relevance

- Plan data analysis strategies that minimize the risk of false discovery and over fitting of classification models
- Develop solid statistical assumptions
- Apply conservative feature selection and statistical cross-validation within a sample set

Develop Data Analysis Strategies Before Acquiring Data

With help from the biostatistician, plan the data analysis workflow and methods you will use before acquiring data. Take into account the clinical question being asked, the study type, and the methods and success criteria being used for the project. Planning the analysis strategy ahead of time ensures that your experimental design generates the amount and types of data you need to perform the types of analysis you have planned.

Use Appropriate Samples, Controls, and Standards

Select samples and controls using the general principles defined in bulletin 5642. For each phase of a project:

- Include one control sample on each array for quality control (QC sample). The total number of QC sample spots can be reduced with very large sample sets
- Use calibration and reference samples that are good models for the experimental samples. For example, dilute protein standards in a sample that is similar to that being analyzed (serum, plasma, urine, etc.), ensuring calibrant concentrations are within the concentration range expected for the experimental samples

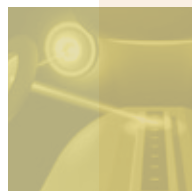
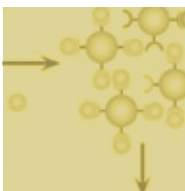
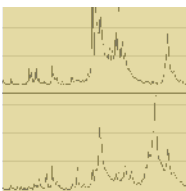
Optimize Acquisition Protocols Before Acquiring Data

The ProteinChip SELDI instrument uses acquisition protocols to acquire data from a spot or portion of a spot on a ProteinChip array. For best results:

- For every experimental condition, test the protocols on a pooled sample before collecting data from study samples
- Optimize the settings for the low- and high-mass ranges separately to ensure sufficient coverage of the entire mass range
- As a starting point, use the collection parameters for general protein profiling recommended in bulletin 5642
- Evaluate spectra during optimization for the maximum number of well-resolved and sharp peaks
- After optimizing laser energy settings, maintain constant data collection settings for all samples associated with that condition

Part II: Data Organization

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Data Organization

The quality of data analysis relies not only on the quality of the spectral data, but on the quality of annotations and folder organization as well. ProteinChip® data manager software stores spectral data in project folders and subfolders created during data collection and analysis. It uses the folder organization as the basis for later peak cluster creation. This chapter describes the steps required to organize the data for analysis with ProteinChip data manager software.

Ensure Proper Annotation of Spectra

Sample annotations are the descriptive information associated with a sample and its analysis. Annotations may include patient information (for example, patient name or sample number, disease group, disease staging or severity, drug treatment, age, sex) as well as sample preparation details (for example, fraction number, matrix, stringency of binding and wash buffers). Annotations are used for grouping spectra and as the basis for distinguishing sample groups during determination of biomarker candidates and development of classification algorithms.

Workflow

1. Compile sample annotations prior to the start of experimental work.
2. If using ProteinChip data manager software, Enterprise Edition, use the **Virtual Notebook** feature to easily link sample annotations with the spectra.
3. Review the annotations to ensure they are complete before proceeding with data analysis.

Objectives

All data analyses rely on the information contained in sample annotations, so it is vital that all annotations are complete, correct, and unambiguous. In addition to unique identifiers, annotations should include clear, complete, and consistent descriptions of the samples and sample treatment procedures that were used.

More About Annotation

Annotations entered in the **Virtual Notebook** before data collection are automatically saved as metadata with the spectral data. They can be updated after data collection by synchronizing the **Virtual Notebook** or by manually editing the spectrum table. Annotations cannot be updated after peak clustering is performed.

Group Spectra Into Folders

ProteinChip data manager software stores data in project folders and subfolders created during data collection and analysis. Spectra should be organized into separate folders within a project folder: base the organization on differences in experimental conditions, and separate analytical control and standard spectra from experimental spectra. ProteinChip data manager software allows you to easily organize spectra by selecting a series of fields for automatic segregation of spectra into conditions. See the ProteinChip Data Manager Software Operation Manual for details.

Workflow

1. Separate and group spectra according to the experimental conditions under which they were obtained.
2. Separate spectra collected from mass calibration standards and from QC and process control samples into different folders than those containing experimental spectra.

Objectives

ProteinChip data manager software generates clusters and performs biomarker analysis for all of the spectra within a folder. The list of biomarker candidates is specific for each experimental condition. To ensure comparison of relevant spectra and to avoid a potential source of analytical bias, be sure spectra are separated appropriately by condition.

More About Folders and Conditions

In ProteinChip data manager software, conditions are defined as the combination of experimental variables that generate different profiles from the same sample. Experimental variables include the following:

- Fraction number
- Array chemistry
- Stringency of binding buffers
- Stringency of wash buffers
- Matrix
- Mass optimization range
- Laser energy

Different combinations of these experimental variables are used to maximize proteome coverage and increase the potential for protein biomarker discovery.

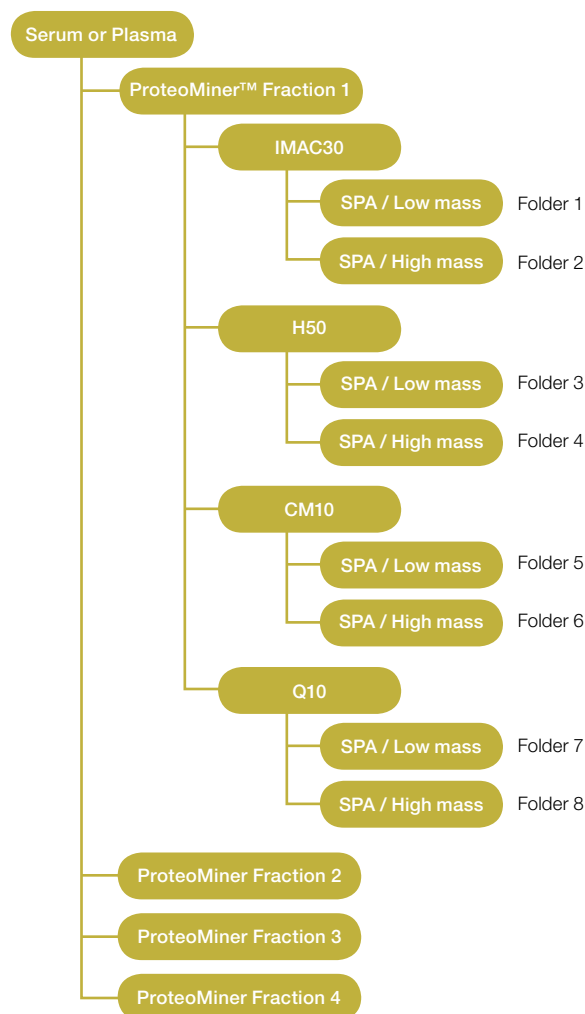


Fig. 5. Grouping spectra according to condition. In a typical study, a condition may consist of the fraction number, array type, matrix, and mass optimization range used. In this example, a serum or plasma sample was fractionated into 4 ProteoMiner fractions, which were profiled on 4 array chemistries (with a single binding and wash stringency per chemistry), using sinapinic acid (SPA) as the matrix. Data were acquired at 2 laser energies (optimized for both the low- and high-mass ranges). The data are grouped into 32 separate analysis folders: (4 fractions) x (4 array chemistries) x (1 matrix) x (2 laser energies). In this example, the 8 folders for fraction 1 are shown in detail; similar folders apply for the other 3 fractions.

For example, an experimental design might include sample fractionation followed by profiling on several ProteinChip array chemistries, use of different wash conditions, application of different matrix molecules, and data collection using different instrument settings. Each combination of experimental variables constitutes a unique condition within which one can compare relative peak intensities (Figure 5).

Table 1. Controls, standards, and reference samples used in ProteinChip SELDI analysis.

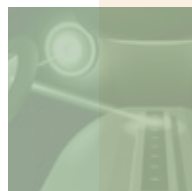
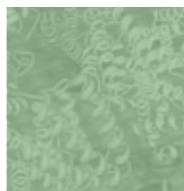
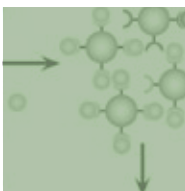
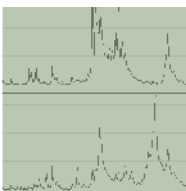
Sample	Description
Experimental control	Healthy (normal) control samples or patient control samples. Healthy controls are samples from individuals with no apparent diseases or disorders. Patient control samples may include disease, treatment, or time-course controls selected to best address the clinical question under investigation
Mass calibration standard	Molecular weight standards, such as the ProteinChip all-in-one peptide standard or ProteinChip all-in-one protein standard II, spotted on a single array and used to collect mass calibration spectra
Quality control (QC)/ process control reference sample	Well-characterized, pooled sample that is processed alongside experimental samples. Use spectra to: 1) monitor instrument performance during routine use, 2) compare protein profiles to historical reference profiles, and 3) calculate median CV for peak intensities as a quantitative measure of reproducibility. Use a pool of study samples for laser energy optimization

Do not make comparisons across different conditions, as differences associated with sample preparation and data collection (analytical bias) will mask relevant protein changes. Organizing the spectra according to condition enables independent analysis of data sets having identical histories of sample treatment, array processing, and data collection; this maximizes proteome coverage while minimizing analytical bias.

For each experiment, include appropriate experimental controls. In addition, include one sample on each array for quality control (QC sample) and use calibration and reference samples that are good models for the experimental samples (Table 1). Group the spectra obtained from these samples into separate folders, and use them to calculate median or pooled coefficients of variation (CVs) of peak intensities and to monitor the quality and reproducibility of SELDI experiments.

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Mass Calibration

The ProteinChip® SELDI system measures how long it takes for a molecule to travel the length of the flight tube and reports this time as TOF data. ProteinChip data manager software uses a calibration equation to convert TOF data into mass data in daltons. This chapter describes the mass calibration process, which involves two steps: internal calibration (generation of a calibration equation) and external calibration (application of the calibration equation to experimental samples). Updating and applying the calibration equation improves the mass accuracy of spectra.

Create the Calibration Equation Using Calibration Standards

Internal calibration generates the calibration equation using data obtained from mass calibration standards. These standards contain peptides or proteins of known mass (for example, the ProteinChip all-in-one peptide and protein standards). The calibration equation is created by assigning known masses to corresponding peaks of the calibration spectrum (Figure 6 and Table 2).

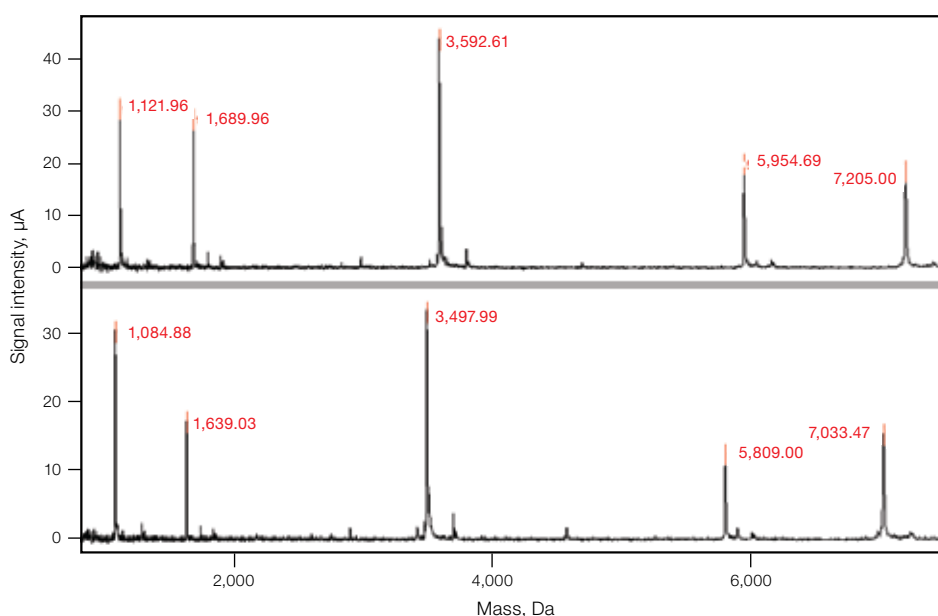


Fig. 6. Effect of calibration. Calibration spectrum before (top) and after (bottom) internal calibration. Table 2 shows the expected and resulting masses for the peaks in the spectrum before and after calibration.

Workflow

1. Manually label the calibrant peaks in a recently acquired peptide or protein calibration spectrum.
2. Use the **Internal Calibration** feature to create the calibration equation.
3. Evaluate the residuals of the calibration equation in the **Internal Calibration** dialog.

Objectives

Internal calibration should generate a calibration equation with low residuals (errors), which are measured in parts per million (ppm). Low residuals indicate better fit of the equation to the calibration data, and equations with low residuals are better suited to sample data. High residuals indicate problems with peak labeling.

- With the ProteinChip all-in-one peptide standard (catalog #C10-00005), errors of <300 ppm are acceptable and errors of <100 ppm are achievable
- With the ProteinChip all-in-one protein standard II (catalog #C10-00007), errors are higher than those of the peptides and errors of <1,000 ppm are acceptable

If high residuals are observed, adjust the peaks chosen as calibrants and ensure that:

- Appropriate peaks were selected
- Peaks were assigned the appropriate mass
- All calibrant peaks were labeled at the apex (zoom in on each peak during labeling)

Recommendations

Calibrant Preparation and Data Collection

- Prepare calibrants separately for each matrix molecule used
- Collect data for all matrix molecules, mass ranges, and laser energies used for experimental samples

Table 2. Improvement in mass accuracy after calibration. Listed are the expected and observed masses for the peaks shown in Figure 6. Note the improvement in mass accuracy after calibration.

Peptide	Average Mass (Da)		
	Expected	Before Calibration	After Calibration
Arg8-vasopressin	1,084.25	1,121.96	1,084.88
Human somatostatin	1,637.90	1,689.96	1,639.03
Bovine insulin, β -chain	3,495.94	3,592.61	3,497.99
Human insulin	5,807.65	5,954.69	5,809.00
Hirudin, recombinant	7,033.61	7,205.00	7,033.47

- Use the data collection protocols that will be used for the experimental samples
- Acquire calibration spectra on a regular basis (for example, as part of weekly instrument operational qualification), even if there is no immediate need to calibrate, and especially after instrument maintenance. New equations can be created and applied as long as the calibration spectra exist
- Use the same calibration equation for the entire experiment unless instrument maintenance is performed in the middle of the experiment or the experiment is performed over a long period of time (for example, more than one month)

Calibration Equation

- Use at least 4 calibrants that span the mass range of interest
- Be wary of byproducts, such as sodium adducts or peak oxidation products, which can generate extra peaks in the calibration spectrum. Exclude such peaks from further analysis
- Since the number of parameters (2 or 3) determines the degrees of freedom used to create the equation: for 4 or more calibrants, use the 3-parameter fit, and for 3 calibrants, use the 2-parameter fit
- Use the **Weighted calibration** option, which allows more even error distribution across peak masses, so high-mass peaks do not overshadow low-mass peaks
- Confirm that calibration spectra were internally calibrated (indicated by a green icon in the spectrum table)
- Save equations using names that facilitate their retrieval and application to other spectra

More About Internal Calibration

ProteinChip data manager software defines internal calibration as the process used to generate the calibration equation that is used for external calibration of experimental spectra. Another, more common use of the term refers to the direct use of peptides of known mass within an experimental spectrum for mass determinations of unknown species. To add peptides to a sample for this type of internal calibration, add them to the experimental sample prior to data collection, keeping in mind that they may interfere with the detection of other peaks.

Improve Mass Accuracy Through External Calibration

The software automatically applies a default external calibration equation to all spectra during data collection. Update this equation on a regular basis to maximize the mass accuracy of the experimental data.

Workflow

1. Select an internally calibrated spectrum and use the **Copy Calibration** function to copy the calibration equation.
2. Select all of the experimental spectra in a folder and use the **Apply Calibration** function.
3. Evaluate the effects of calibration.

Objectives

External calibration should improve the mass accuracy of experimental spectra, which facilitates subsequent data analysis steps such as peak clustering. Properly calibrated data also enable comparison of biomarker candidates from different experimental conditions or across different studies.

Recommendations

- Update the default calibration equation before performing other data processing steps
- Use an equation generated from a calibration spectrum that was collected using: 1) calibrants that encompass the mass range of interest, and 2) data collection settings that are similar to those used with the experimental samples (focus mass and laser energy are the most critical collection settings)
- Confirm that spectra have been externally calibrated (indicated by a yellow icon in the spectrum table). Examine the equation parameters to check whether the values are from a generated or default equation
- To review the calibration information for a spectrum (for example, the name of the external spectrum used or the calibration equation name and details), select the **Peaks & Calibration** tab from the **Spectrum Properties** menu

Processing Spectral Data

During data collection, ProteinChip data manager software uses default parameters for processing spectral data. These processing steps include baseline subtraction, filtering, and noise estimation. Often, the default processing parameters may not be appropriate for every sample type or condition. This chapter describes how to optimize these processing parameters and reprocess the spectra prior to peak clustering.

General recommendations for processing spectra include:

- For optimization of processing parameters, the largest differences exist for data collected using different matrix molecules and mass optimization ranges
- For each folder, optimize the processing parameters for a few representative spectra, and then apply the changes to all other spectra in the folder

Adjust Peak Intensity Calculations Using Baseline Subtraction

Mass spectrometers cannot always deliver a flat baseline. To compensate, ProteinChip data manager software applies a baseline subtraction algorithm to each spectrum. This algorithm subtracts the noise contributions to calculate more accurate peak intensities.

Workflow

1. Examine the shape of the baseline in several reference or representative spectra in each folder. Toggle the **Subtract Baseline** function to visualize the baseline.
2. If necessary, adjust the values for the **Fitting Width** and **Smoothing** settings.
3. Evaluate effects of any change on the appearance of the spectrum, and work to generate a baseline that is consistent with the shape of the spectrum.
4. For optimum reproducibility, apply the same baseline subtraction settings to all spectra within a condition.

Objectives

The baseline of a spectrum should be relatively flat, and the peaks should be sharp and well-resolved. Large humps in a spectrum, or the appearance of numerous peaks with negative intensity indicate that the baseline subtraction algorithm may require adjustment (Figure 7).

With the calculated baseline visible, the baseline should follow the base of all peaks in the spectrum. Gaps between the calculated baseline and spectrum indicate that the fitting width is too large; overlaps indicate that the fitting width is too small (Figure 7).

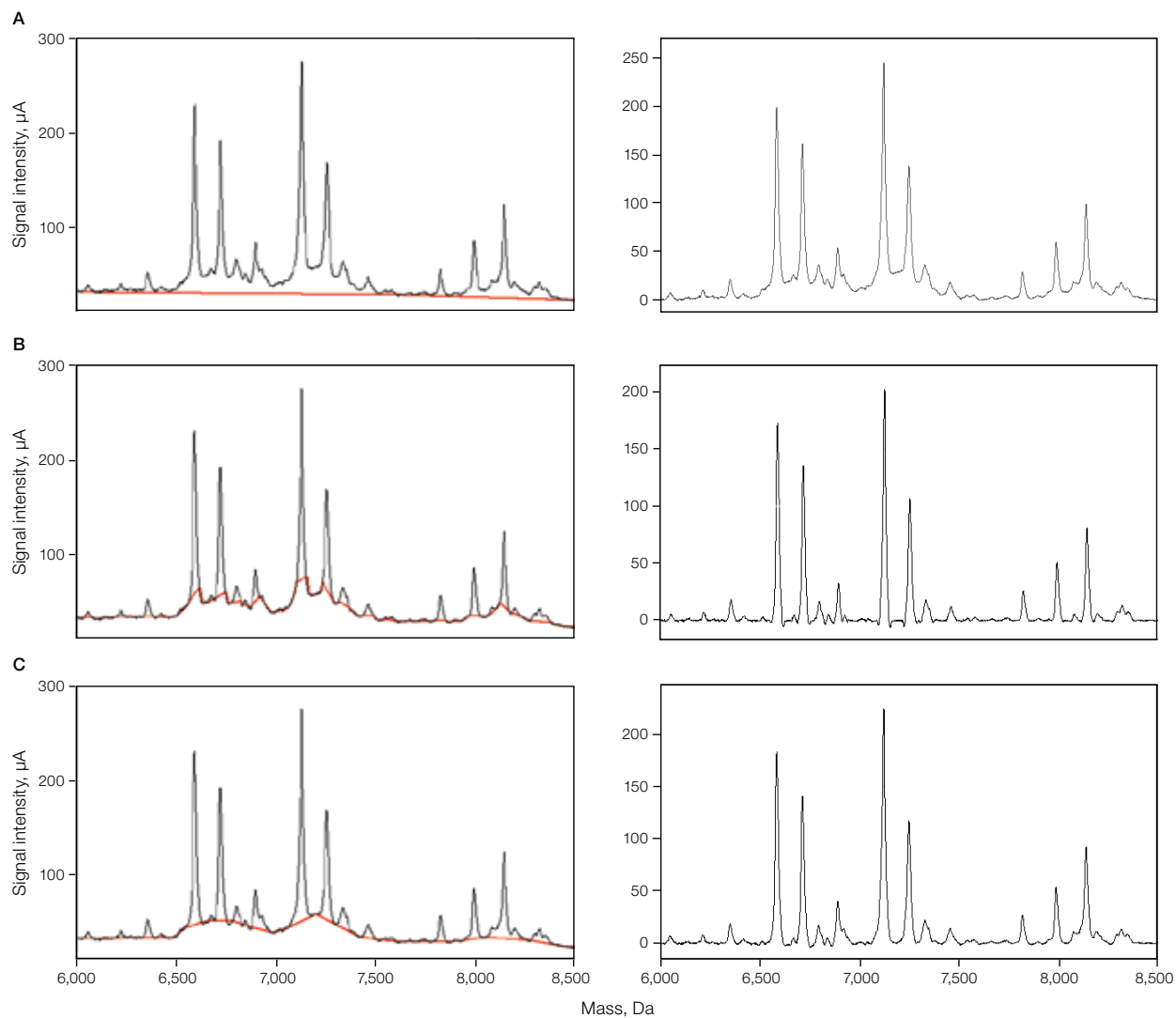


Fig. 7. Effects of baseline subtraction and fitting width. In all panels, spectra are shown before (left) and after (right) baseline subtraction, and the calculated baseline is shown in red for contrast (the software displays the baseline in dark gray). **A**, the large gap between the baseline and spectrum indicates the fitting width setting is too high. **B**, the appearance of numerous peaks with negative intensity values indicates the fitting width setting is too low. **C**, the baseline follows the shape of the spectrum, and the spectrum after baseline subtraction is flat with sharp peaks, indicating the fitting width has been optimized for the spectrum.

Recommendations

Setting the Fitting Width

- Use the same value for all spectra within a condition
- If the default baseline is not appropriate for a particular sample, disable the **Automatic** setting and examine the spectra using different values for the fitting width. Use values 10–20 times the expected peak width as starting points for optimization
- If you are not able to identify a fitting width that works well across an entire mass range, make a copy of the data and analyze each mass range as a separate condition

Using the Smoothing Function

- Use the same settings for all spectra in a condition and set the **Smoothing** function to be left on
- For spectra optimized for the low-mass range, the default **Window** value of 25 points for smoothing is appropriate
- For spectra optimized for the high-mass range, set the **Window** value to 10 points
- If peak height is overestimated due to high noise values, increase the **Window** value for smoothing by increments of 5 from the recommended starting values

More About Baseline Subtraction

The software calculates peak height, or signal intensity, as the distance from the top of a peak to the calculated baseline. Ideally, the baseline intensity value would be close to zero; however, owing to chemical and electronic noise, this is rarely the case.

The software applies a baseline subtraction algorithm to each spectrum when calculating peak areas and intensities. Adjusting the baseline subtraction parameters in the software allows some control over the shape of the baseline and yields better representations of relative peak heights. Due to the physics of mass spectrometry and the heterogeneity of high-mass proteins, baselines must be adjusted differently for different mass ranges.

The main parameter that controls baseline shape is the fitting width, which defaults to an automatic value. Depending on the mass range, data collection settings, matrix molecule, and other variables, however, this setting can be optimized.

Smoothing, which is often confused with filtering, is an option used to compensate for small negative spikes from very noisy data to ensure that the baseline is not underestimated. Adjusting the smoothing values controls the influence of noise variability on the peak intensity.

Reduce Noise by Filter Adjustment

Filtering removes noise from a spectrum to improve the signal-to-noise ratio, particularly in the high-mass range. This improves peak detection. Changing the filtering setting is optional; the default settings are effective for most situations.

Workflow

1. Specify the value for noise filtering. In most cases, the default **Average** filter with a setting of 0.2 times expected peak width is appropriate.
2. Analyze the effects of any changes to the filter settings on the spectrum. If resolution is compromised, revert to the original settings.
3. Apply the same filter setting to all spectra in a condition.

Objectives

The objective of filtering is to decrease the amount of noise in the spectrum (Figure 8), but overfiltering may result in loss of resolution. The amount of filtering required for a spectrum depends mainly on the mass range being analyzed.

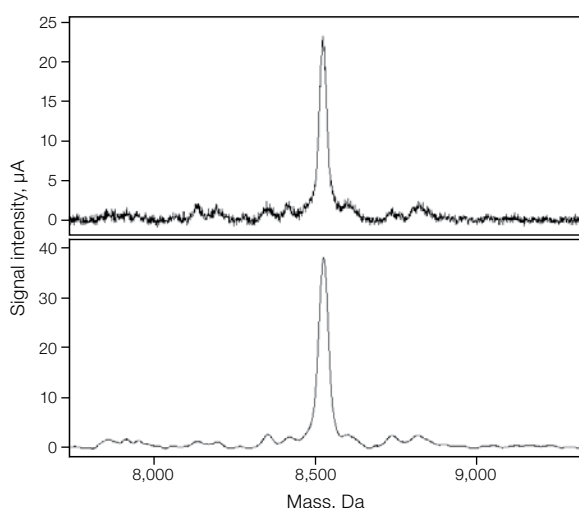


Fig. 8. Effects of filtering. A spectrum without filtering (top) exhibits significantly more noise than a spectrum with filtering (bottom).

Recommendations

- Leave the filter setting enabled (on); otherwise, the signal-to-noise ratios for high-mass peaks, which tend to be broad, will be insufficient for detection
- Use the default **Average** filter setting of 0.2 times expected peak width, which is suitable in most cases
- To generate higher signal-to-noise ratios for improved peak detection when analyzing data in the high-mass range (>20 kD), raise the filtering setting to 1.0 times expected peak width for maximum filtering
- Examine the effects of using the **Savitsky-Golay** filter when looking at peaks below 2,500 Da

More About Filter Adjustment

By default, ProteinChip data manager software applies the **Average** filter and usually generates excellent results without any adjustment. Excessive filter widths can distort peaks and reduce resolution. In fact, adjustments may slightly change peak shape and, therefore, the location of the peak apex, which can affect estimations of peak masses.

The **Average** filter calculates a variable-width moving average that is optimized to work over the entire mass range and is used to decrease the overall amplitude of noise. Possible units of the filter width are times expected peak width, Da (daltons), or data points. Using the **times expected peak width** option causes the width of the filter to vary with mass, becoming wider at higher masses where peaks are wider. Allowable values for this unit are 0.1–1.0; 1.0 is preferred for data in the high-mass range.

In some cases, such as peptide identification when the peaks are narrow, the **Savitsky-Golay** filter is more appropriate. This filter preserves peak shapes of narrow peaks as compared to the **Average** filter. Use this filter if the main peaks of interest are below 2,500 Da. Only daltons or data points are allowed as width units when using this filter.

Improve the Signal-to-Noise Ratio by Setting the Noise Range

The noise of a spectrum represents the average intensity of the data points in areas where there are no peaks. Noise is a measure of the variation in the signal, not of the signal itself. Adjusting the noise calculation improves signal-to-noise calculations, which improves peak detection.

Workflow

1. Set the starting mass for noise calculation to the matrix attenuation value (specified during data collection) plus $\geq 10\%$.
2. Analyze any effects of this change on the peak counts from automatic peak detection (no visible change is made to the spectrum).
3. Apply the optimized noise setting to all spectra in a condition.

Objectives

The goal of setting an appropriate range for defining the spectrum noise is to improve the accuracy and reproducibility of automatic peak detection by excluding the signal generated by the matrix or by matrix attenuation.

Recommendations

- For the **Measure Noise to** setting, enable **End at End** (that is, the end of the spectrum)
- Exclude peaks in the matrix attenuation range from analysis
- Apply the same settings to all spectra in a condition (assuming they have the same matrix attenuation value)

More About Setting the Noise

Automatic peak detection by ProteinChip data manager software depends on the ability of the software to discriminate the peaks created from peptides or proteins striking the detector from those generated by random electronic and chemical noise. For peak determination, the software uses the ratio of the intensity of a peak to the noise in that region. If this signal-to-noise ratio exceeds the set threshold value, the software detects and labels the peak (to override the threshold, use manual detection and labeling). The accuracy of noise measurement, therefore, influences the signal-to-noise ratio of labeled peaks.

A number of factors influence noise, including the presence of many peaks and the effect of matrix attenuation. Since the noise varies along the m/z (mass) axis, the noise algorithm divides the axis into a predefined number of equal segments when estimating the noise. The noise at a location is estimated using the standard error of a linear regression of the data points in a segment surrounding that location.

Processing Conditions

Perform normalization and alignment for all the spectra within each folder (condition) at the same time. Normalization and alignment help to maximize reproducibility and mass precision to enable determination of the most robust peak clusters. This chapter describes how to perform normalization and alignment.

Standardize Intensities by Normalization

Total ion current (TIC) normalization standardizes the intensities of a set of spectra to compensate for any spectrum-to-spectrum variations (Figure 9). Variations may be caused by minor differences in total protein concentration, sample preparation, or data collection. Within a folder, select spectra to normalize by highlighting them in the spectrum information table. Generally, all spectra in the folder should be selected for TIC normalization at one time.

Workflow

1. Use the **Normalize Spectra** function to normalize all spectra within a condition by their TIC. Specify the minimum starting mass, the ending mass, and whether to use an external normalization coefficient.
2. Generate a histogram displaying normalization factors (Figure 10). Inspect spectra with normalization factors outside the normal distribution to assess the protein profiles, and eliminate spectra of poor quality prior to statistical analysis.
3. If outlying spectra are removed from the analysis, repeat step 1 (not required if normalizing to an external coefficient).

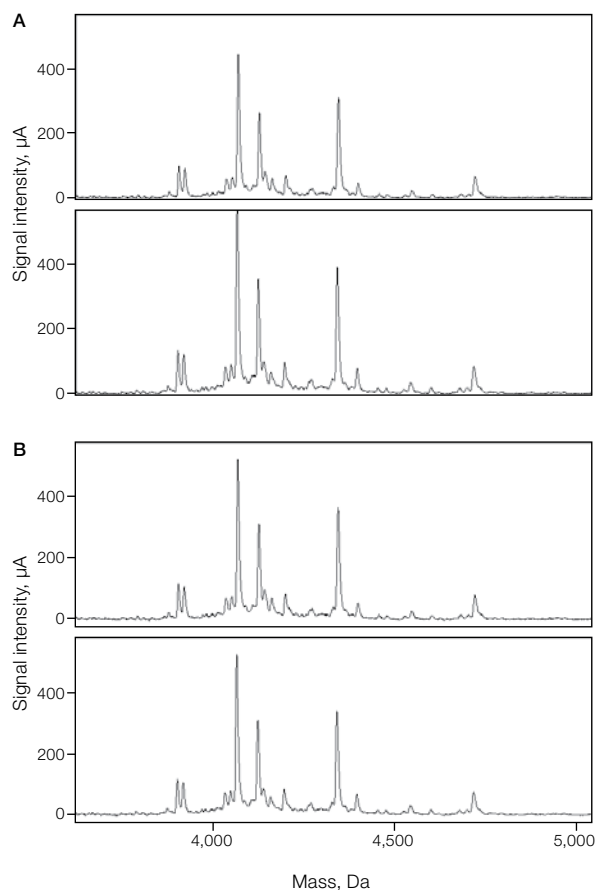


Fig. 9. Effects of TIC normalization. Two spectra before (A) and after (B) normalization. The difference in peak heights is reduced following normalization, improving reproducibility.

Objectives

Normalization improves spectrum-to-spectrum reproducibility. Normalized spectra should exhibit more similar peak heights (Figure 9). Normalization also generates normalization factors, which can be used to identify spectra of poor quality (Figure 10).

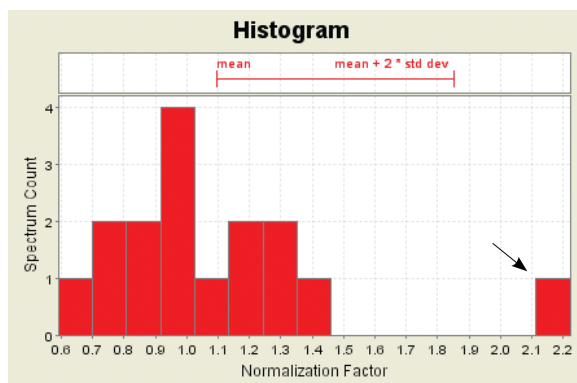


Fig. 10. Histogram of normalization factors. The histogram can be used to identify spectra with normalization factors that are outside the acceptable range. The arrow in this example indicates the outlier.

Recommendations

- Perform normalization after any changes are made to the baseline calculation, as the baseline has a direct effect on the TIC
 - Perform normalization by TIC separately on each condition — each condition (combination of fraction, array type, wash condition, and data collection settings) represents a unique subset of the total protein content and should be normalized independently
 - For conditions covering a large mass range, copy the spectra into different folders and normalize the low-and high-mass spectra independently
 - Perform normalization by TIC on all groups within a condition together to avoid group bias
 - Adjust the starting mass for normalization for different laser energies and detector blanking settings to be equal to or greater than the lower cutoff for noise definition. Specifically, start from the matrix attenuation setting used for data collection plus 10% to exclude matrix noise from normalization
- In many cases, such as those requiring comparisons of peak intensities across experiments, specify an external normalization coefficient that is the same for both data sets (for example, an external normalization coefficient of 1.0 or a value close to the original coefficient). When using an external coefficient, the resulting peak intensities may not correlate well with the actual peak intensity, but the relative differences within one study and between studies are preserved

More About TIC Normalization

The TIC is the sum of the peak intensities across the length of a spectrum (the area under the spectrum). The normalization process calculates the average TIC from all selected spectra to generate a normalization coefficient. The coefficient is then used to calculate normalization factors for each spectrum, which are then used to adjust the intensity scales: values of >1 indicate the spectra have been scaled up, and values of <1 indicate they have been scaled down. Once a spectrum is normalized, the normalized peak intensities are used for downstream biomarker analysis.

This method assumes that the TIC contribution of peaks that change between experimental groups (treatment or disease, for example) is insignificant compared to the overall protein profiles. In cases where the total protein content or TIC differs significantly between relevant groups (for example, in antibody capture experiments), TIC normalization may not be appropriate.

Normalization factors can be used to identify spectra of poor quality. Spectra with extreme normalization factors are usually generated by samples containing air bubbles or particulate matter and should be removed from analysis. If all replicates fail to meet these criteria, there may be a problem with the integrity of the sample; in such cases, remove the samples from further analysis.

Improve Mass Precision by Spectrum Alignment

Spectrum alignment reduces mass variations of peaks within a group of similar spectra. After alignment, the same peaks across all spectra have higher mass precision. This step is optional and often unnecessary for data collected using instruments that have been properly calibrated. It is recommended when clustering is problematic, for example when a shoulder is labeled instead of a peak apex.

Workflow

1. If mass shifts occur between spots or arrays, label at least 5 well-resolved peaks in the mass range of interest using the **Centroid All** tool to ensure inclusion of all spectra.
2. Select the reference spectrum, which serves as the standard mass template.
3. Evaluate the effect of alignment (Figure 11), and either save or revert the results. If reverted, the calibration equation of the spectra is not changed. If saved, the calibration equations are updated to the newly aligned equations.

Objectives

The goal of alignment is to increase the mass precision of peaks across all spectra within a condition. Following alignment, aligned peaks are plotted in the **Alignment results** window (Figure 11). The window also shows the **pooled %CV**, which should be lower following spectrum alignment.

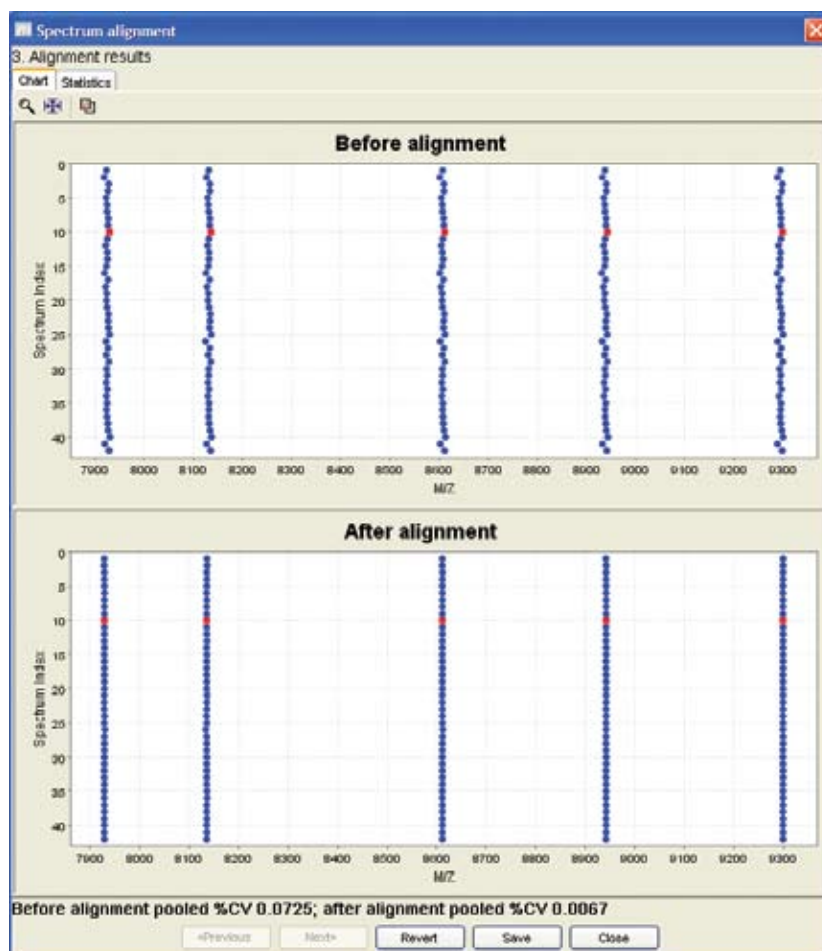


Fig. 11. Effects of spectrum alignment.

This example shows 5 peaks across 42 spectra, with each dot representing a peak label. Note the improvement in pooled %CV after alignment. Alignment was performed with a high number of peaks selected in the mass range of interest (<10,000 Da).

Recommendations

General Recommendations

- If a peak is not represented in the alignment plot (this may occur if the peak is outside the default 0.3% mass range), widen the mass window until it appears. This effect is more common in the high-mass range due to broadening of peaks
- If a peak has a low signal-to-noise ratio, it may not appear in the alignment plot. Decrease the minimum signal-to-noise ratio setting until it appears; however, avoid using very small or spurious peaks in alignment
- Confirm that spectra have been aligned (indicated by a blue calibration icon in the spectrum table)

Manual Peak Selection

- Select at least 5 well-resolved peaks in the mass range of interest to use as peaks for alignment
- Use peaks that occur in all spectra, and concentrate these peaks in the region of interest
- Peaks outside of the alignment range may not be calibrated properly. Include 1 or 2 peaks at the beginning and end of the spectra to minimize this effect

Automatic Peak Selection

- Use of automatic peak detection for the selection of peaks for alignment is not recommended due to the high probability of overlapping and missing peaks
- Label all peaks within a spectrum over a given signal-to-noise ratio in both the reference spectrum and in the spectra to be aligned as an alternative to manual peak selection
- For automatic peak detection, a high signal-to-noise setting is recommended (>20), coupled with manual inspection or use of the **Minimum signal/noise ratio** option to decrease dependence on small peaks

Reference Spectrum Selection

The masses of the reference spectrum peaks become the standards against which spectra are aligned. The reference spectrum should:

- Represent the peaks occurring in the experimental spectra; for example, use a spectrum from a well-characterized pool of samples such as a QC sample
- Contain a large number of well-resolved peaks
- Have undergone internal calibration, where possible, to ensure maximum mass accuracy

More About Spectrum Alignment

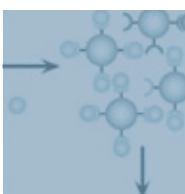
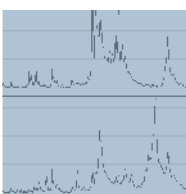
Spectrum alignment is similar to mass calibration and uses the same algorithms. The **Spectrum alignment** tool uses labeled peaks within a reference spectrum to create a new calibration equation for each spectrum. However, unlike calibration, the masses of the peaks are not known; rather, they are correlated with peak masses from the reference spectrum.

If a reference peak does not match any peak in a sample spectrum, that reference peak is not used for that spectrum. Since alignment involves recalibration, the original calibration equation is overwritten. This results in higher precision between the spectra. Accuracy is determined by the mass accuracy of the peaks in the reference spectrum.

If mass shifts occur between spots A–H in reproducible patterns over several arrays, use the spot correction process to adjust for these shifts. These corrections are valid only for newly acquired data and cannot be applied to existing data.

Part IV: Generation and Analysis of Peak Clusters

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Generation of Peak Clusters

Peak clusters are groups of peaks of similar mass that are treated as the same substance (protein or peptide, for example) across multiple spectra. ProteinChip® data manager software uses statistical analyses of peak clusters to uncover biomarker candidates that exhibit different expression levels between sample groups. This chapter describes how to create and edit peak clusters.

Associate Peaks From Multiple Spectra Into Peak Clusters

Peak clustering labels peaks across all the spectra in a folder and then groups them together based on matches between their calculated masses. The result is a list of masses, or peak clusters,¹ that contain peaks of similar mass from each spectrum with associated intensities for each sample. This list is interpreted as a list of peptides and proteins with their associated expression levels.

Workflow

1. Ensure that all data processing steps have been performed and that all annotations have been added or updated. After creating peak clusters, sample annotations and data processing parameters cannot be changed without undoing the peak clusters.
2. Delete all existing peak labels.
3. Create peak clusters using the **Create EDM** (expression difference mapping) feature by specifying parameters for peak detection and cluster completion.

¹ These peak clusters should not be confused with the results of statistical methods of clustering, such as hierarchical clustering (discussed later in this guide).

Objectives

The main objective of peak clustering is to create, within a condition, a list of all the peaks and their corresponding intensities across all the spectra. Ideally, peak clusters contain all the peaks from the spectra within a condition, with each spectrum contributing one peak to each cluster. If peaks are absent from a spectrum, baseline intensities are estimated and included for statistical calculations. The cluster table lists all the clusters and provides statistics for intensity and mass.

Recommendations

General Recommendations

- For initial biomarker screening, delete manually labeled peaks prior to clustering to ensure all peaks meet the specified thresholds
- Since a folder should represent a single experimental condition, do not specify a condition in the first page of the **Cluster Wizard**; leave the field blank
- Analyze control samples separately to measure analytical reproducibility
- During later validation or assay phases of a project, cluster only manually detected peaks for first-pass detection to ensure labeling of only the relevant peaks. Alternatively, save a list of peak clusters and automatically apply them to other groups of spectra

Parameters

The main parameters for clustering are the sensitivity setting for **Peak Detection** and the mass window setting for **Cluster Completion**. To control the sensitivity with which peaks are detected for clustering, specify the signal-to-noise ratio, valley depth, and minimum percentage of spectra in which to require peak detection. To control the mass window within which the software looks for peaks to include in a peak cluster, specify the peak width or percentage of mass.

- During the discovery phase of a biomarker research project, use automatic first-pass peak detection. As a guideline for serum samples, use initial settings of 5 times the signal-to-noise ratio and a valley depth of 3 for first-pass peak detection
- Use a second pass of detection that is lower than the first pass (usually set to 2 for valley depth and 2 for peak signal-to-noise). This setting is optional and may be left unchecked
- Use the **Preserve all first pass peaks** feature to ensure that first-pass peaks used in the creation of a cluster are retained in the cluster, even if they fall outside of the final cluster mass window once all peaks are labeled
- For experimental samples, select a minimum peak threshold percentage that represents roughly half the number of samples in the smallest group, but that is at least 10–20%. For example, if there is a 50:50 distribution of the sample groups, choose a setting of 25%; if the experiment contains 3 groups of equal numbers, use a minimum peak threshold of 15%
- Define the mass window for peak clustering as approximately 0.3% of the peak mass (low-mass range) and approximately 2% of the peak mass (high-mass range). The larger mass window for the high-mass proteins accounts for broadening of peaks due to numerous possible isoforms. Alternatively, define the mass window using peak width settings to automatically adjust for this peak broadening
- During addition of estimated peaks, use the **auto centroid** option (the **at cluster center** option is included for backward software compatibility)

More About Peak Clustering

Peak clustering is divided into three operations:

- First-pass peak detection — determination of the total number of clusters to be created by either automatic peak detection (the default) or manual detection and clustering of user-defined peaks only
- Second-pass peak detection (optional) — addition of smaller peaks to the clusters created in the first pass. For this phase, use automatic peak detection and adjust the sensitivity of the second pass by lowering the detection threshold to detect smaller peaks that were not found in the first pass. This step may be disabled and addition of estimated peaks used instead to complete the cluster
- Addition of estimated peaks — addition of estimated peaks to spectra that do not have a peak meeting the minimum signal-to-noise setting. This ensures that every spectrum is represented in the cluster. The two options for adding estimated peaks are **at cluster center**, which places all estimated peaks at the midpoint of the cluster, and **auto centroid**, which places an estimated peak using a centroid operation

When optimizing the automatic peak detection settings for clustering, the objective is to accurately label peaks without labeling spurious peaks generated by chemical or electronic noise. Some overlabeling can be tolerated, however, since spurious peaks usually fail to demonstrate statistical significance and would be excluded during statistical analysis or quality control of candidate biomarkers.

Detection Threshold

The detection threshold is a combination of the signal-to-noise ratio and valley depth values used by the software for peak detection (see sidebar). Higher detection thresholds yield fewer peak labels and, therefore, a smaller pool of biomarker candidates. Using lower detection thresholds yields more candidates for analysis, but may include more clusters of lower peak quality.

Minimum Peak Threshold Percentage

The minimum peak threshold percentage is the percentage of the total number of spectra in which a peak must be found for it to comprise a cluster. ProteinChip data manager software uses this value to minimize the number of clusters that are based

on spurious peaks (peaks that only occur in a small number of spectra). For example, the default value of 20% requires that a peak appear in at least 20% of all spectra before a cluster is created. A value of 100% requires that a peak is found in every spectrum, and a setting of 0% (not recommended for most applications) results in a cluster being formed for every peak of unique mass. The appropriate value depends on several factors, such as the number of different sample groups in the experiment and the variability between samples as discussed above.

Cluster Mass Window

The cluster mass window defines the mass width of the cluster. Clusters must be wide enough to accommodate any differences in mass between spectra, but small enough to prevent two or more

distinct peaks of similar mass from being treated as part of the same cluster. Spectrum alignment may be performed prior to clustering, especially if there are obvious mass shifts between spectra (as may occur when analyzing spectra collected on different instruments or separated by time).

Once the cluster centers are established during first-pass detection, the cluster width is specified as one of the following:

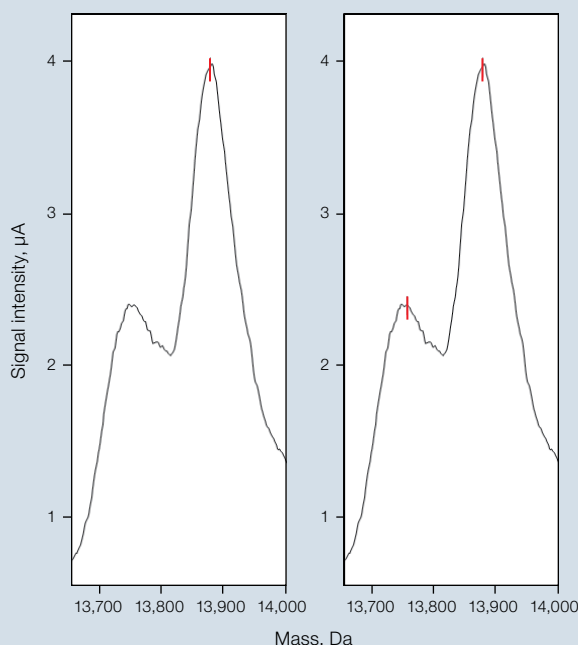
- Percentage of mass — used when limiting the mass window width; the preferred selection when using manually defined peak clusters
- Peak width — dynamically adjusts the mass window based on the width of the peak; useful for initial discovery

Automatic Peak Detection — Valley Depth and Peak Height

The automatic peak detection algorithm of ProteinChip data manager software identifies peaks by locating changes in intensity that are greater than specified thresholds. In some instances, the areas of adjacent peaks overlap, generating peaks with shoulders. While the entirety of these peaks are not visible, they should still be evaluated as potential biomarkers. These shoulders, however, can be difficult to label using a single signal-to-noise threshold. To enhance detection of shoulder peaks, ProteinChip data manager software uses two signal-to-noise thresholds:

- Valley depth — peak signal measured from the nearest valley, divided by the noise measured at that mass
- Peak height — peak signal measured from the baseline, divided by the noise measured at that mass; must be greater than or equal to the valley depth setting

The software transiently labels all peaks that fulfill the valley depth requirement. This labels resolved peaks, shoulder peaks, and peaks of low intensity. The peak height setting filters out low-intensity peaks, retaining only prominent and shoulder peaks (see figure).



Effects of peak detection settings. Left, spectrum after peak detection using a valley depth of 5 (disabling peak height), which does not label the shoulder peak. Right, same spectrum after peak detection using a valley depth of 3 and a peak height of 5, which labels the shoulder correctly.

Adjust Peaks Within Clusters Using Cluster Editing

To ensure accurate statistical analysis, the output of peak clustering must be complete and precise. Mislabeling of peaks may occur if peaks are very broad, contain shoulders, or if the mass window for peak clustering was too wide. Peak labels that are not automatically placed at the appropriate peak centroid can be manually adjusted to the correct location using the cluster editing process within ProteinChip data manager software (Figure 12). This allows you to shift peak labels within a cluster, without having to repeat the cluster generation process for the entire condition.

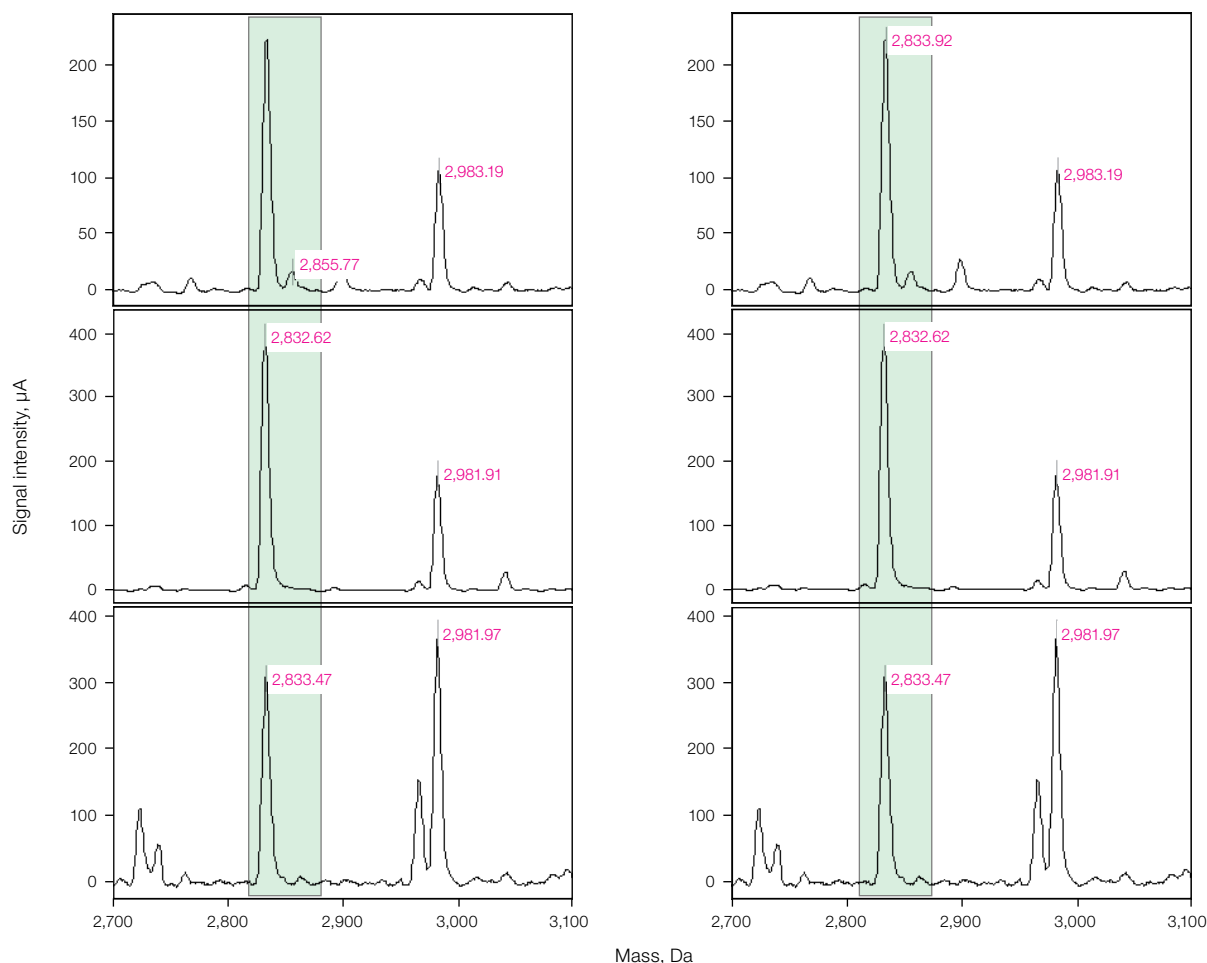


Fig. 12. Effect of cluster editing. Left, spectra before cluster editing, showing a peak labeled incorrectly (2,855.77 Da) in the top spectrum. Right, the same spectra after cluster editing, where the peak label was moved to the correct centroid at 2,833.92 Da.

Workflow

1. Use the **Plot Spectra in Cluster** function from the **Clusters** tab to view spectra.
2. Examine the spectra to identify peak labels that are not placed appropriately, as compared to other spectra in that peak cluster.
3. Use the **Edit Cluster** or **Move Peak(s)** cursor to move a peak label from one position on the spectrum to another.
4. Update the statistics in the cluster table.

Objectives

Cluster editing resolves any problems caused by mislabeling of peaks and results in more accurate statistical analysis of biomarker candidates.

Recommendations

- Cluster editing is manually intensive and can be time-consuming. If many clusters require editing and relabeling, perform cluster editing after the initial statistical screening step to focus on biomarker candidates of interest. Alternatively, repeat clustering using different parameters such as mass window
- After editing clusters, recalculate statistics for edited clusters by clicking **Update edited cluster statistics** to update the cluster table
- Note that clusters cannot be edited if sample replicates have been selected for P value calculation. If necessary, temporarily deselect replicates using **Calculate P-values**, edit clusters as above, and recalculate statistics with appropriate replicates selected

Selection of Candidate Biomarkers

Statistical analyses of peak clusters identify clusters that exhibit significant differences in intensity values (expression levels) between clinically relevant sample groups. In biomarker research, univariate statistical methods are used to consider differences in the expression levels of one biomarker candidate at a time. This chapter describes the selection and evaluation of single candidate biomarkers using univariate analyses.

Select Biomarker Candidates Using P Values

In ProteinChip data manager software, the P value calculation rapidly screens peak cluster data for biomarker candidates.

Workflow

1. Use the **Calculate P-values** function to specify the grouping criterion that differentiates the experimental groups within a condition.
2. With replicates, specify the field that contains the replicate information (usually the **Sample name** field).
3. Examine the **p-value** column for peak clusters that display statistically significant differences between groups.
4. Assess the quality of the underlying peaks using the **Plot Spectra in Cluster** feature. Manually inspect peak clusters with good P values for peak quality (good signal-to-noise ratio, resolution, accurate peak labeling, etc.) using the **Spectrum Viewer** and the 3 scatter plots (Figure 13).

Objectives

No formal P value threshold applies for all experiments. The numbers of samples and candidate biomarkers, as well as other statistical principles, such as false discovery rate, all play a major role in determining the appropriate P value for a given experiment. However:

- P values of <0.05 are generally considered significant
- P values of <0.01 are a reasonable starting point for more robust markers

After locating biomarker candidates by sorting for low P values, assess the quality of the underlying peaks using the **Plot Spectra in Cluster** feature. In this step, differentiate clusters consisting of high-quality peaks from low-quality clusters consisting of spurious peaks. Exclude low-quality clusters from further analysis, because they represent noise and exhibit high variability in their intensities, making them poor biomarkers.

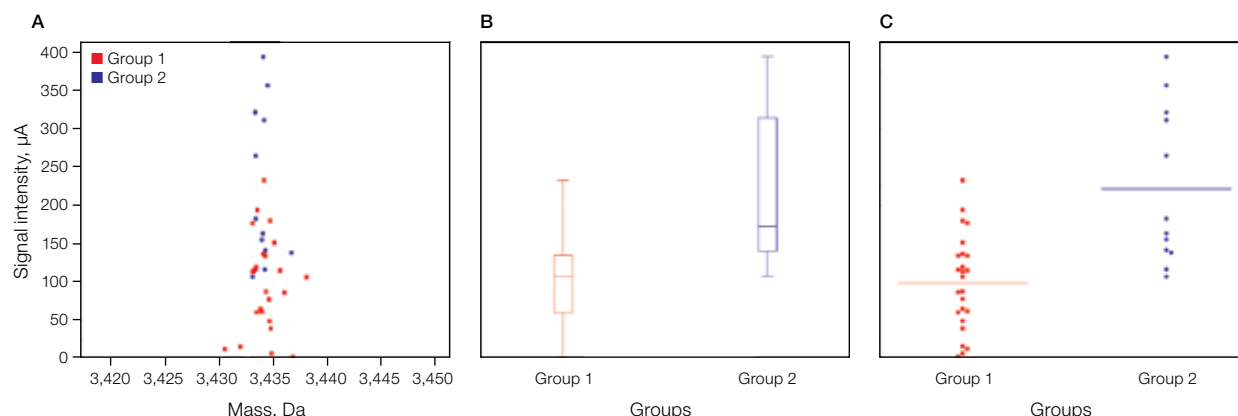


Fig. 13. Graphic representations of cluster data. ProteinChip data manager software represents cluster data using three different plot options. **A**, m/z mass scatter plot displaying the intensity and mass values for a selected cluster. **B**, group box and whiskers plot indicating 0, 25th, 50th, 75th, and 100th percentiles for sample groups within a cluster. **C**, group scatter plot displaying average group intensities as horizontal lines.

Recommendations

- Average replicates before calculating P values
- Sort the cluster table by P values to simplify location of low P values
- If the number of estimated peaks in the cluster table is greater than 75% of the total number of peaks in the cluster, this may indicate a cluster of poor quality, or one that is close to the current detection limit. It may also indicate a peak that is only present in one patient group, which could lead to a good biomarker candidate
- Evaluate the fold change and distribution of intensities during quality control of significant peaks to ensure selection of the most robust markers. The P value tests implemented in the software are rank-based tests; only the relative rankings of the intensities are taken into account, not the actual intensity difference
- Multiple (>2) sample groups can be merged into smaller sample groups using the **Merge groups** function in the **P-Value Wizard**. For example, analyze male and female, treated and control subjects separately as four groups, or combine and analyze them as controls (male + female) vs. treated (male + female), without repeating the clustering process
- The grouping chosen in the **P-Value Wizard** is also used to create receiver-operating characteristic (ROC) plots, and in hierarchical clustering and principal component analysis (PCA). If new groupings are required for these analyses, use the **P-Value Wizard** to change the groups

More About P Values

The P value calculation tests the null hypothesis that the median peak intensities of the groups being compared are equal. The P value can be interpreted as, “I’ve rejected the null hypothesis that the groups have the same median, with an X% chance that I am wrong,” where $X = P \text{ value} \times 100\%$. Thus, low P values indicate that there may be a significant difference in expression levels for a particular protein (peak cluster) in the groups examined.

ProteinChip data manager software uses nonparametric tests and takes into account only the ranking of the intensities during the P value calculation. Nonparametric tests do not assume a normal distribution of ProteinChip SELDI data. The Mann-Whitney test is used for two-group comparisons, the Kruskal-Wallis test for three or more groups, and the Wilcoxon signed-rank test for paired data.

If possible, use the assistance of a biostatistician in formalizing an appropriate P value threshold. The ProteinChip SELDI system has the ability to evaluate large numbers of clusters in a single experiment, and the biostatistician may implement a test of multiple comparison (for example, Bonferroni or false discovery rate) to determine the P value threshold. In general, 0.01 is a good starting point if there are multiple clusters with P values below 0.05. Use other statistical tools, such as ROC curves (described in the next chapter), in conjunction with P values in the evaluation of biomarker candidates.

Measure Biomarker Sensitivity and Specificity Using ROC Curves

For two-group comparisons, use ROC curves to compare the sensitivity and specificity of a biomarker candidate at different cutoff values for peak intensity. The sensitivity and specificity for a group (for example, control vs. treated or normal vs. disease) is related to the amount of overlap in protein expression levels, and this overlap is quantifiable using the area under the ROC curve (AUC).

Workflow

1. Use the **P-Value Wizard** to automatically construct ROC plots and calculate AUC values. AUC is automatically calculated when P values are calculated, assuming only two groups are specified.
2. Examine the **ROC area** column of the cluster table to identify peak clusters with good AUC values.

Objectives

The ROC plot provides an evaluation of the ability of a biomarker candidate to indicate a disease, response to treatment, or other difference in sample groups.

The area under the plot, the AUC, provides a quantitative measurement of this potential: as the AUC approaches a value of 1.0, it indicates higher sensitivity and specificity (Figure 14).

AUC is calculated with respect to one of the groups, which is called the positive group. Since this designation is arbitrary in a biomarker discovery experiment, the AUC for a good biomarker candidate may also be close to 0.0. Changing the positive group inverts the value to calculate the true AUC.

- AUC values close to 1.0 and 0.0 indicate good biomarker candidates
- A value of 0.5 indicates that the protein expression levels of both groups completely overlap

Examples of ROC curves for good and poor biomarker candidates, and their relationships to box-and-whisker plots, are shown in Figure 14.

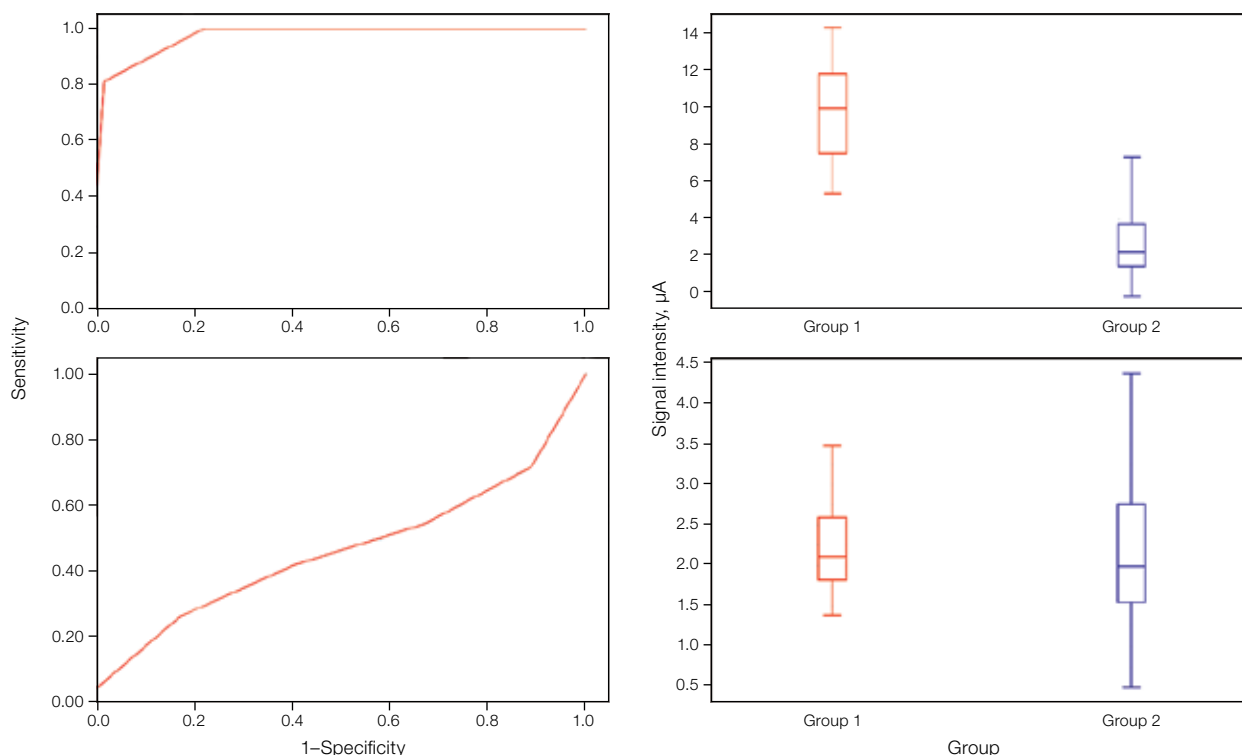


Fig. 14. Examples of ROC plots (left) and corresponding box-and-whisker plots (right). Pairs of plots for candidate biomarkers exhibiting good (top) and poor (bottom) sensitivity and specificity. Note the high degree of overlap in the bottom box-and-whiskers plot.

Recommendations

- Since AUC values close to 1.0 or 0.0 can indicate viable biomarker candidates, sort the **ROC area** column by ascending and then by descending values in the table to see both extremes
- If the AUC is below 0.5, toggle the **Positive group** in the ROC plot to invert the graphical representation and display the AUC in a standard format
- Use the **Merge groups** function of the **P-Value Wizard** to create a two-group comparison from multiple groups
- The grouping chosen in the **P-Value Wizard** is also used by ROC, hierarchical clustering, and PCA. If new groupings are required for these analyses, use the **P-Value Wizard** to change the groups

More About ROC Curves

To construct an ROC curve, the software first computes the true positive ratio (sensitivity) and the false positive ratio (1–specificity) for a small set of cutoff values of protein expression level. The group to which each subject belongs is known, and the software calculates the true positive ratio by dividing the number of diseased subjects, for example, that are considered positive by the total number of actual diseased subjects. The false positive ratio is obtained in similar fashion.

- All subjects with peak intensities greater than the cutoff value are considered positive (for example, diseased)
- All subjects with peak intensities less than the cutoff value are considered negative (for example, healthy)

These calculations are performed at a range of cutoff values (ProteinChip data manager software uses six to eight intervals), the results are plotted, and the area under the plotted curve is calculated as the AUC.

The AUC provides a quantitative measurement of the accuracy of a diagnostic test. Whereas an AUC of 0.5 indicates no separation between groups, an AUC of 1.0 indicates 100% sensitivity and specificity. AUC is normally reported as a value of >0.5, which indicates the peak is upregulated in the selected positive group. During screening for biomarker candidates, also examine values of <0.5, which represent peaks that are downregulated in the selected positive group but still show good sensitivity and specificity.

Note that ROC is a two-group comparison. If P values are calculated for a condition that contains more than two groups, the values displayed under **ROC area** appear as **NaN**, indicating AUC cannot be calculated.

Assessment of Sample Relationships

Once candidate biomarkers are selected, unsupervised multivariate analyses such as hierarchical clustering and principal component analysis (PCA) can be used to reveal both desired (disease diagnosis, drug treatment efficacy, adverse drug effect) and undesired (preanalytical or analytical bias) associations within the data. These unsupervised techniques determine associations of peak clusters using only the intensity values within the cluster data, without bias from the properties of the samples that generated them. This chapter describes the application of hierarchical clustering and PCA for examining the entire protein expression profile and revealing sample relationships within an experiment.

Examine Sample Hierarchical Clustering Patterns

Hierarchical clustering is an unsupervised, exploratory learning method of multivariate analysis that groups multidimensional data by similarity. This method generates heat maps and dendrograms (Figure 15), widely used visualizations of high-dimensional data such as microarray data in gene expression studies.

Workflow

1. Select the sample group for examination using the **P-Value Wizard**.
2. Launch the **Heat Map** feature to visualize peak clusters as green and red points with variable shades based on relative expression level.
3. Use the dendrogram to create a clustered heat map that illustrates sample relationships.
4. Examine the clustered heat map for correlations between patterns of samples and the grouping criterion.

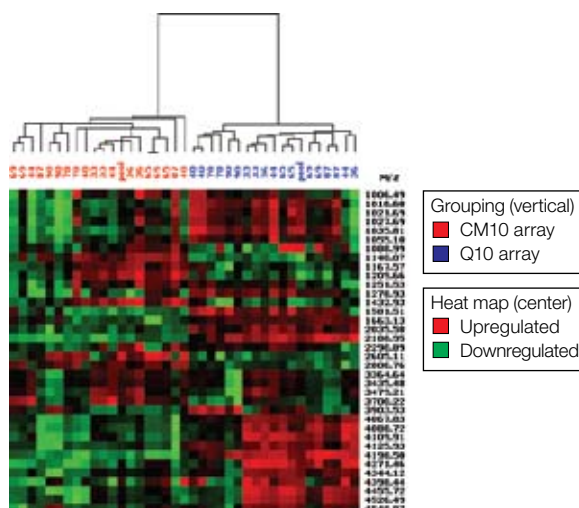


Fig. 15. Example of a heat map and dendrogram of SELDI data. In the heat map, red indicates high protein expression compared to the mean intensity value, and green indicates low expression. Black indicates no change from the mean. In this example, samples are grouped by array type, illustrating the difference in expression profiles created by each surface chemistry.

Objectives

The objective of unsupervised hierarchical clustering is to observe groupings of spectra based on their complete expression profiles.

In the clustered heat map, the arrangement of samples can indicate accurate clinical grouping, unknown confounding factors, or possible subgroups with common clinical characteristics (for example, cancer metastasis vs. no metastasis subgroups may cluster together within the group previously defined as cancer).

The proximity of spectra and the branching patterns in the dendrogram reflect sample similarity. Spectra that are in close proximity within a branch exhibit similar expression profiles: the branch length reflects the degree of the relationship, with shorter branches indicating closer relationships. Options allow you to display the relationships and similarities between expression profiles of samples (columns) and between expression levels of peak clusters (rows).

Recommendations

- Exclude peak clusters from the heat map using objective metrics, such as poor peak quality. Do not select clusters based on their P value or AUC for unsupervised clustering applications
- The grouping chosen in the **P-Value Wizard** is also used by ROC, hierarchical clustering, and PCA. If new groupings are required for these analyses, use the **P-Value Wizard** to change the groups
- When examining an experiment for sample bias, choose grouping criteria that represent experimental variables that may be sources of bias or confounding factors
- Correlate branching patterns with clinical characteristics by selecting those characteristics as the grouping criteria
- Hierarchical clustering can be used with or without averaging of replicates. If replicates are not averaged, examine the sample arrangement to see if the replicates are near each other with close correlation
- The two heat maps displayed by the software are copies of each other. Change the sizing as desired to zoom into regions of interest

More About Hierarchical Clustering

In hierarchical clustering, as in PCA, the relative positions of samples in the dendrogram are determined solely by the intensity levels of the peaks. They are independent of any sample characteristics. This is in contrast to P value or ROC curve statistics, in which the grouping criterion selected for a sample (for example, control or treated) directly affects the resulting value. As a result, any criterion that correlates with a particular arrangement of samples may have a causal relationship to the sample profiles.

The heat map is a graphical representation of the entire data array in grid form: each column represents a spectrum (sample), each row represents a peak cluster, and each cell represents a particular peak, which is color coded according to its expression level compared to the mean of the other peaks in the sample. ProteinChip data manager software creates heat maps within a single or across multiple conditions; note, however, that multicondition heat maps require consistent naming of sample across all conditions.

Use hierarchical clustering when multiple peak clusters exhibit differences. Single-peak differences are not easily visualized on heat maps. For information about using supervised hierarchical clustering of a limited number of candidate clusters selected by good P value or AUC, see the sidebar at the end of this chapter.

Visualize Sample Similarity Using PCA

PCA is another unsupervised multivariate analysis method used to visualize multidimensional data by mathematically reducing dimensionality of data into components that approximate the original multidimensional data. In the case of ProteinChip SELDI data, each cluster of peak intensities is considered its own dimension. ProteinChip data manager software uses PCA to visualize spectra in two- and three-dimensional graphs that illustrate relationships between spectra based on their expression profiles (Figure 16).

Workflow

1. Select a grouping criterion using the **P-Value Wizard**.
2. Launch the **PCA** feature to automatically generate PCA plots.
3. Examine the distribution of the samples in the PCA plots to identify possible associations with the grouping criterion.

Objectives

The objective of unsupervised PCA is to visualize the degree to which the samples in the experiment are similar. Unlike other visualizations in ProteinChip data manager software, where a dot on a plot represents a peak, each dot on a PCA plot represents the expression profile of an entire spectrum. Simply put, the closer dots on a PCA plot, the more similar the expression profiles. If the colors representing a particular sample group appear close together and separate from others in a PCA plot, those samples are closely related and exhibit differences from other sample groups. If the colors are interspersed, there is no correlation between protein expression and that sample group (Figure 16).

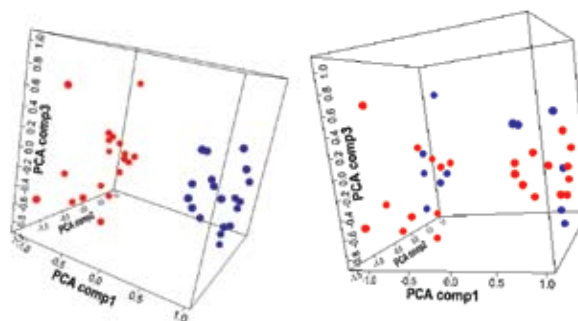


Fig. 16. Example of PCA plots. Left, plot generated using disease state as the grouping criterion. Right, plot generated using buffer lot as the grouping criterion. In each plot, colors reflect sample group, and the relative positions of each color reflect the expression profile. Note the higher correlation and separation of sample types when disease state is used as the grouping criterion. PCA comp = PCA component.

Recommendations

- At the **PCA parameters** dialog, select the **Use correlation matrix** option and deselect the **Use log transformed data** function
- Use clinical grouping criteria to search for variables that illustrate desired correlations to protein expression, such as disease state or effect of drug treatment. For biomarker discovery, a separation of colored dots with the selected criteria is the goal
- Use different analytical or preanalytical grouping criteria to search for variables that illustrate unwanted correlations to protein expression, such as collection site or date prepared. Interspersed colored dots reflect the absence of the selected confounding factor
- Use the 3-dimensional plot to visualize 3 principal components at one time

More About PCA

SELDI data sets consist of multiple variables, and it is difficult to visualize all of the data points at the same time. For example, with intensity values collected on 50 clustered peaks from 100 spectra, there are 100 observations and 50 variables. It is difficult, and not very instructive, to plot the data points in a 50-dimensional space.

One simple solution is to use scatter plots to project multivariate data into a two-dimensional space defined by just two of the variables. Though this allows examination of pairwise relationships between variables, such simple projections may conceal more complicated relationships. PCA enables detection of such relationships.

While the complete definition of PCA is fairly long, it may be summarized as a technique used to visualize multidimensional data by approximating the data along different vectors. These vectors are defined by weighted linear combinations of variables, termed principal components, and they form the axes in the multidimensional space.

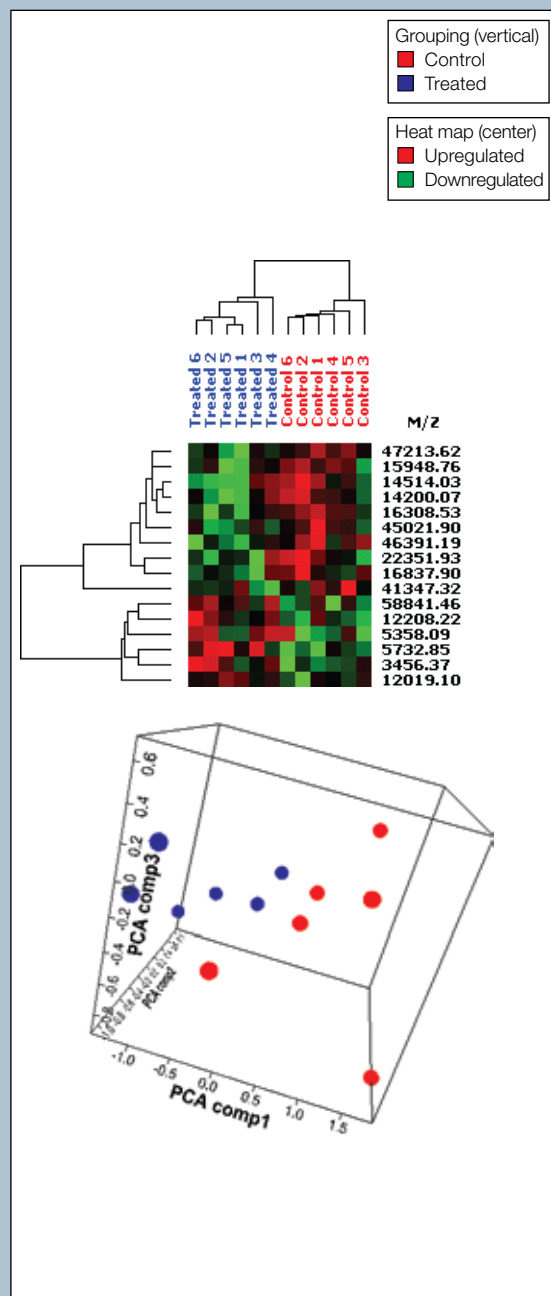
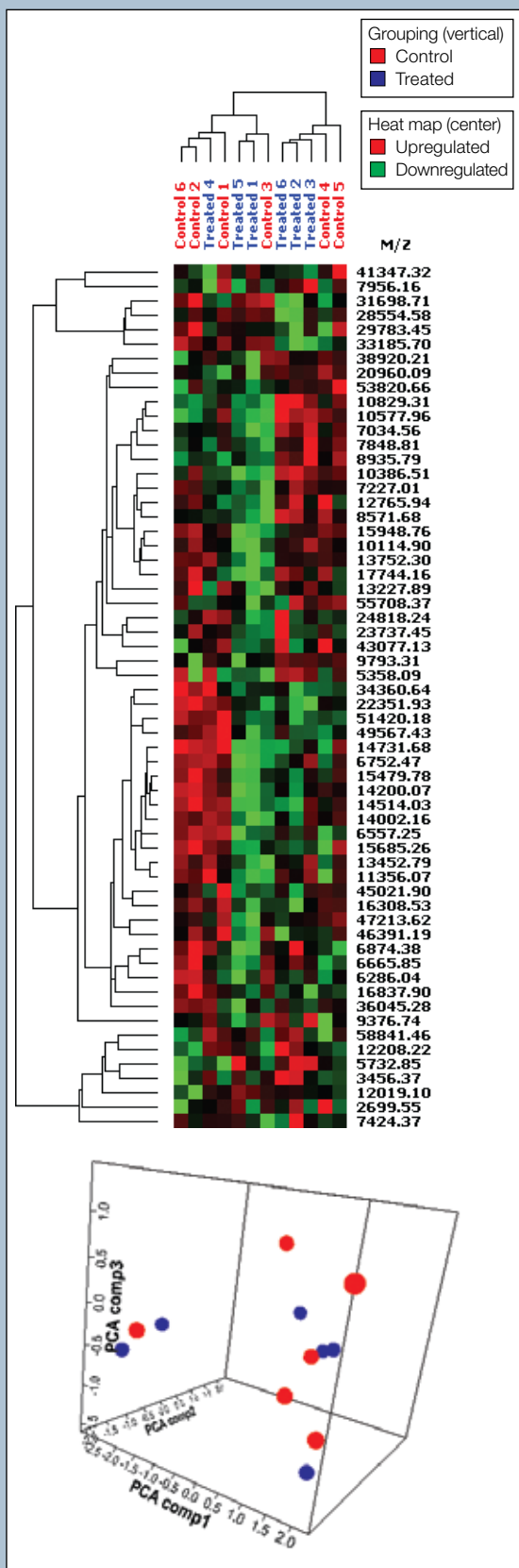
Principal components are ranked based on how closely they represent a multidimensional space. The component names then are based on those rankings (for example, PCA comp1, PCA comp2, PCA comp3, etc.). The plots default to plotting the first three components; however, any three components may be chosen. Limit picking components to the first six or seven components from the **Pick components** feature. The correlation of a principal component to the original data drops off rapidly as the principal component number increases, as illustrated in the **Scree plot**, which plots relevance (% variance explained) against component number.

The distribution of points for a set of peak clusters is always the same for a given PCA plot. Changing the grouping criteria only changes the coloring of the dots.

Supervised Hierarchical Clustering and PCA

Often, the individual biomarkers identified by univariate analysis are insufficient to completely distinguish between two sample groups, as observed in overlap of peak intensities on group scatter plots. Though ProteinChip data manager software cannot create supervised classification algorithms (see the Going Beyond ProteinChip Data Manager Software chapter), it can be used for visualization of supervised hierarchical clustering and PCA.

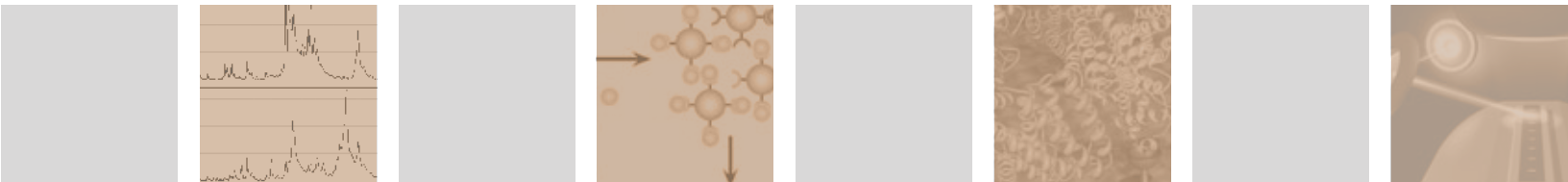
In these methods, heat maps or PCA plots are generated using a subset of biomarker candidates selected for their good P values or AUC values. The resulting dendrograms or PCA plots generally illustrate better separation of the sample groups than any of the individual biomarker candidates. These visualizations demonstrate the benefits of combining multiple markers for better classification.



Unsupervised and supervised hierarchical clustering and PCA. Left, unsupervised heat map (top) and PCA plot (bottom) generated using the entire expression profile. Right, supervised heat map and PCA plot generated using biomarker candidates selected following univariate statistical analysis. Effective differentiation of control and treated samples was achieved by supervised hierarchical clustering and PCA by combining a set of good biomarker candidates, none of which individually could completely separate both groups.

Part V: Going Beyond ProteinChip® Data Manager Software

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 - Validate Candidate Biomarkers42



Going Beyond ProteinChip Data Manager Software

The output of the workflow presented in this guide is a list of candidate biomarkers that should be subjected to additional statistical analyses and experimental validation. In addition, biomarker candidates are often identified and more specific assays developed for improved quantitation. This chapter outlines the types of further multivariate analyses to consider, how to export data from ProteinChip data manager software, and various aspects of designing a validation study.

Develop and Test Classification Algorithms

The workflow presented thus far generates a list of biomarker candidates through univariate statistical analysis. Each of the candidate biomarkers, when used alone, may not distinguish completely between clinical groupings (for example, a diseased patient from a healthy one), as the expression levels of single biomarkers generally overlap between diagnostic groups. This is particularly true in human trials, in which patient-to-patient variability is high. In these cases, multivariate analysis methods outside those available in ProteinChip data manager software are required to analyze patterns of protein expression and develop algorithms to improve sample classification.

The final step of data analysis using ProteinChip data manager software, outlined in the previous chapter, involves hierarchical clustering and PCA. These two unsupervised multivariate techniques uncover desired or undesired relationships between samples: desired relationships can confirm diagnostic groupings or lead to additional understanding of the clinical questions under investigation, and undesired relationships can indicate possible confounding factors within the study, such as sources of analytical or preanalytical bias.

In contrast, supervised multivariate techniques, which are limited in ProteinChip data manager software, are used to develop and test classification algorithms. Supervised multivariate classification techniques incorporate multiple sample variables to create algorithms that define known clinical groupings within a set of data. A second set of data from the same experiment is withheld from algorithm development and is used later, in a blinded fashion, to test the performance of the algorithm. When unblinded, the predictive performance of the test data set should approximate the predictive performance of the original data set used to define the algorithm.

A number of algorithms are available commercially, including partial least squares, maximum likelihood, K nearest neighbor, support vector machines, and classification tree analyses. These techniques can be used with peak cluster data exported from ProteinChip data manager software (see below).

ProteinChip pattern analysis software (also known as biomarker patterns software) can be used to perform supervised multivariate analysis and generate an easy-to-interpret decision tree that uses a small panel of markers with defined splitting rules for classification. Following multivariate analysis, several decision-tree models from the discovery phase are typically selected for further testing and refinement in the validation phase of the research project.

Export Data for Further Analysis

To use the supervised multivariate analysis methods mentioned above, export the data from ProteinChip data manager software using one of the following methods (consult the ProteinChip Data Manager Software Operation Manual for more information):

- Export cluster intensities — creates a table of the intensities within each peak cluster (the protein expression levels from each sample). Each cluster forms a column, and each sample forms a different row. These data are often imported into third-party statistical software for further analysis
- Export peak information — creates a table of information containing the characteristics of each peak labeled within the selected spectra. This information is often used to perform CV calculations to evaluate reproducibility
- Export processed data — exports all of the m/z and intensity coordinates for the data points that make up a spectrum. With this information, it is possible to plot the processed spectrum in other graphing programs that apply additional or alternative types of data processing or peak detection

If possible, collaborate with biostatisticians throughout study design and during the complex phases of statistical data analysis, especially when moving to multivariate analysis.

Validate Candidate Biomarkers

To ensure the predictive value of candidate biomarkers, assess their validity against a larger, more heterogeneous population. This involves analyzing a large number of samples using a reduced set of experimental conditions (for example, those yielding the biomarker candidates). Inclusion of a more diverse population facilitates the selection of the most robust markers.

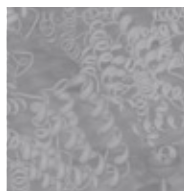
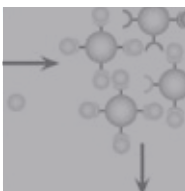
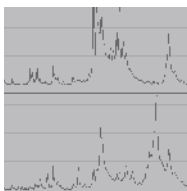
The design of the validation phase of a biomarker research project can vary depending on the study setting: it may repeat and confirm the findings from the discovery phase on a larger sample set or it may explore different clinical variables that may affect the validity of the markers on a large population and, ultimately, the clinical utility of the biomarkers themselves.

Validation often follows the discovery of candidate biomarkers and precedes development and application of clinical assays. The throughput and data analysis capabilities of the ProteinChip SELDI system allow rapid and efficient validation of candidate biomarkers from large numbers of samples, which can increase the statistical confidence in a biomarker. However, comprehensive clinical validation studies often demand prospective sample collection and require extensive numbers of samples and periods of time. In addition, purification and identification of biomarker candidates is crucial and may be performed immediately following discovery or after one or more validation studies.

For more information about how to design a biomarker research project, and for more details about planning SELDI biomarker discovery and validation experiments, refer to bulletin 5642, Biomarker Discovery Using SELDI Technology: A Guide to Successful Study and Experimental Design.

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Glossary

Array

ProteinChip® array; samples are applied to spots on an array for analysis with the ProteinChip SELDI reader

Area under the ROC curve (AUC)

Quantitative measurement of the accuracy of a diagnostic test. Whereas an AUC of 0.5 indicates no separation between groups, an AUC of 1.0 indicates 100% sensitivity and specificity

Baseline

Calculated line from which peak intensities are measured

Calibrant

Individual component of a calibration standard

Calibration equation

Equation used to convert measured time-of-flight (TOF) data into mass data (daltons); generated by collecting a spectrum for a calibration standard and associating the known molecular weights with appropriate peaks on the calibration spectrum

Calibration spectrum

Spectrum collected from a calibration standard sample and used to generate a calibration equation

Calibration standard

Mixture of peptides or proteins of known mass that are used to generate the calibration spectrum; mass standard, such as the ProteinChip all-in-one peptide standard or ProteinChip all-in-one protein standard II

Centroid

Point at which a line divides a peak into two parts having equal, integrated area. If the peak is symmetric, the centroid is the apex; if asymmetric, the centroid is shifted toward the more slowly falling edge. Molecular mass in SELDI is assigned using a centroiding algorithm

Coefficient of variation (CV)

Quantitative representation of the reproducibility of a measurement, such as peak intensity. Better reproducibility correlates with lower %CV values

Condition

Combination of experimental variables (for example, fraction number, array chemistry, matrix, mass optimization range, laser energy) that generates a unique profile from the same sample. Spectra are grouped into folders according to identical conditions for downstream data processing and analysis

Confounding factor

Extraneous variable in a statistical model that correlates (positively or negatively) with both the dependent and independent variables

Dendrogram

Plot with branches depicting the correlation of columns or rows of data after hierarchical clustering

External calibration

Application of a calibration equation to an experimental spectrum

External normalization coefficient

Manually entered normalization coefficient used to standardize ion currents across multiple experiments

Filtering

Removes noise from a spectrum to improve the signal-to-noise ratio, particularly in the high-mass range, to improve peak detection

Folders

Analogous to folders in Windows software; ProteinChip data manager software folders can contain subfolders, a set of spectra, a Virtual Notebook, or data acquisition protocols

Heat map

Graphical representation of peak expression data: each column of a heat map represents a spectrum, each row represents a peak cluster, and each cell represents a specific peak that is color coded to reflect its expression level

Hierarchical clustering

Unsupervised clustering technique that groups samples or clusters based on overall similarity and displays the results as a dendrogram

High-mass range

Region of a spectrum above 20 kD, where peaks are generally broader due to chemical heterogeneity. High-mass data collection and analysis protocols are generally optimized to maximize sensitivity of detection

Intensity

Peak height, as measured from the calculated baseline to the top of the peak

Internal calibration

Creation of a calibration equation by assigning known masses directly to calibrant peaks within a calibration spectrum

Low-mass range

Region of a spectrum below 20 kD, where peaks are generally sharp and well-resolved. Low-mass range data collection and analysis protocols are generally optimized to maximize peak resolution

Mass accuracy

Proximity of a calculated peak mass to the actual mass of the biological molecule

Mass precision

Proximity of calculated masses of the same peak across multiple spectra

Mass-to-charge ratio (m/z)

Determines the time of flight (TOF) of ions in the ProteinChip reader. A protein with double the charge travels twice as quickly as a protein of the same mass with a single charge

Matrix

Energy absorbing compound applied to array spots to convert laser energy to thermal energy to facilitate desorption and ionization. Two common matrices used for SELDI protein profiling analyses are sinapinic acid (SPA) and α -cyano-4-hydroxycinnamic acid (CHCA)

Matrix attenuation range

Region of a spectrum preceding the matrix attenuation value specified in the data collection protocol. This area is characterized by signal suppression as compared with the rest of the spectrum

Multivariate statistical analysis

Statistical analysis involving evaluation of a combination of peak clusters; for example, hierarchical clustering and principal component analysis

Normalization factor

Multiplication factor used to rescale peak intensities after normalization

P value

Statistical representation of the chance that the relationship being observed is observed by pure chance. For example, a P value of 0.01 means that there is a 1% chance that the relationship between the independent variable and the dependent variable established by the model does not actually exist

Peak

Segment of a spectrum that rises above the noise level, typically with constraints that the maximum height above noise of the segment is greater than a certain threshold. A minimum steepness of the rise and/or fall in intensity may be used as additional constraints when defining a peak. Compounds (peptide, protein, etc.) retained on the array generate peaks at specific molecular masses when analyzed by SELDI-TOF MS

Peak cluster

Group of peaks across multiple spectra that represent the same compound (peptide, protein, etc.)

Principal component analysis (PCA)

Vector-space transform used to reduce multidimensional data sets for analysis and visualization

Receiver-operating characteristic (ROC) curve

Plot of the true positive ratio (sensitivity) and false positive ratio (1-specificity) for a number of cutoff values of the diagnostic variable, represented in SELDI by the peak intensity (see also AUC)

Reference sample

Well-characterized pool of samples that is processed with experimental samples; also known as quality control (QC) or process control sample

Reference spectrum

Data collected from a reference sample and used to monitor instrument performance during routine use and as benchmarks for comparison during data processing and analysis. In spectrum alignment, the spectrum used as the peak mass template

Resolution

Measure related to the ability to resolve, or distinguish, two adjacent peaks that is calculated by dividing the mass of a peak (m) by the full width at half max (Δm)

Sample annotation

Descriptive information associated with a sample that is stored along with spectra as metadata. Annotations are used for sample tracking and for grouping of spectra during analysis

Sensitivity, analytical

Lowest limit of analytical detection; in SELDI, the amount of a protein or peptide required to generate a peak distinguishable from noise

Sensitivity, clinical

Rate of true positive detection; rate at which the test detects disease in a person known to have it; equivalent to number of true disease samples divided by total disease samples

Signal-to-noise ratio

Ratio of the signal intensity of a peak to the calculated value of the local noise

Smoothing

Option used to compensate for small negative spikes in a spectrum to ensure that peak intensity is not overestimated

Specificity

Rate of true negative detection; rate at which the test does not detect disease in a person known to not have it; equivalent to number of true control samples divided by total control samples

Spectrum

Intensity vs. time-of-flight (TOF) data converted into graphical form

Starting mass

Parameter used for noise calculation used to make sure that noise is measured appropriately and avoids possible spectral artifacts in the region of the matrix attenuation range

Time of flight (TOF)

Time it takes for a particle to travel from the ionization source to the detector

Total ion current (TIC)

Average intensity for a mass spectrum

Univariate statistical analysis

Statistical analysis involving evaluation of a single peak cluster at a time; for example, calculation of P values and areas under receiver-operating characteristic curves

Virtual Notebook

Electronic, or "virtual", representation of a laboratory notebook for tracking of sample information, sample preparation details (fraction, array chemistry, binding conditions, matrix), and data collection protocols

Width

Difference in mass (Δm) between the left and right sides of a single isolated peak as measured at 50% of the peak height; also referred to as Full Width at Half Max (FWHM), Δm is used to calculate peak resolution

Workflow Summary

Data Organization

Data Organization

Ensure Proper Annotation of Spectra

1. Compile sample annotations prior to the start of experimental work.
2. If using ProteinChip data manager software, Enterprise Edition, use the **Virtual Notebook** feature to easily link sample annotations with the spectra.
3. Review the annotations to ensure they are complete before proceeding with data analysis.

Group Spectra Into Folders

1. Separate and group spectra according to the experimental conditions under which they were obtained.
2. Separate spectra collected from mass calibration standards and from QC and process control samples into different folders than those containing experimental spectra.

Mass Calibration and Data Processing

Mass Calibration

Create the Calibration Equation Using Calibration Standards

1. Manually label the calibrant peaks in a recently acquired peptide or protein calibration spectrum.
2. Use the **Internal Calibration** feature to create the calibration equation.
3. Evaluate the residuals of the calibration equation in the **Internal Calibration** dialog.

Improve Mass Accuracy Through External Calibration

1. Select an internally calibrated spectrum and use the **Copy Calibration** function to copy the calibration equation.
2. Select all of the experimental spectra in a folder and use the **Apply Calibration** function.
3. Evaluate the effects of calibration.

Processing Spectral Data

Adjust Peak Intensity Calculations Using Baseline Subtraction

1. Examine the shape of the baseline in several reference or representative spectra in each folder. Toggle the **Subtract Baseline** function to visualize the baseline.
2. If necessary, adjust the values for the **Fitting Width** and **Smoothing** settings.
3. Evaluate effects of any change on the appearance of the spectrum, and work to generate a baseline that is consistent with the shape of the spectrum.
4. For optimum reproducibility, apply the same baseline subtraction settings to all spectra within a condition.

Reduce Noise by Filter Adjustment

1. Specify the value for noise filtering. In most cases, the default **Average** filter with a setting of 0.2 times expected peak width is appropriate.
2. Analyze the effects of any changes to the filter settings on the spectrum. If resolution is compromised, revert to the original settings.
3. Apply the same filter setting to all spectra in a condition.

Improve the Signal-to-Noise Ratio by Setting the Noise Range

1. Set the starting mass for noise calculation to the matrix attenuation value (specified during data collection) plus $\geq 10\%$.
2. Analyze any effects of this change on the peak counts from automatic peak detection (no visible change is made to the spectrum).
3. Apply the optimized noise setting to all spectra in a condition.

Processing Conditions

Standardize Intensities by Normalization

1. Use the **Normalize Spectra** function to normalize all spectra within a condition by their TIC. Specify the minimum starting mass, the ending mass, and whether to use an external normalization coefficient.
2. Generate a histogram displaying normalization factors. Inspect spectra with normalization factors outside the normal distribution to assess the protein profiles, and eliminate spectra of poor quality prior to statistical analysis.
3. If outlying spectra are removed from the analysis, repeat step 1 (not required if normalizing to an external coefficient).

Improve Mass Precision by Spectrum Alignment (Optional)

1. If mass shifts occur between spots or arrays, label at least 5 well-resolved peaks in the mass range of interest using the **Centroid All** tool to ensure inclusion of all spectra.
2. Select the reference spectrum, which serves as the standard mass template.
3. Evaluate the effect of alignment, and either save or revert the results.

Generation and Analysis of Peak Clusters

Generation of Peak Clusters

Associate Peaks From Multiple Spectra Into Peak Clusters

1. Ensure that all data processing steps have been performed and that all annotations have been added or updated.
2. Delete all existing peak labels.
3. Create peak clusters using the **Create EDM** (expression difference mapping) feature by specifying parameters for peak detection and cluster completion.

Adjust Peaks Within Clusters Using Cluster Editing

1. Use the **Plot Spectra in Cluster** function from the **Clusters** tab to view spectra.
2. Examine the spectra to identify peak labels that are not placed appropriately, as compared to other spectra in that peak cluster.
3. Use the **Edit Cluster** or **Move Peak(s)** cursor to move a peak label from one position on the spectrum to another.
4. Update the statistics in the cluster table.

Selection of Candidate Biomarkers

Select Biomarker Candidates Using P Values

1. Use the **Calculate P-values** function to specify the grouping criterion that differentiates the experimental groups within a condition.
2. With replicates, specify the field that contains the replicate information (usually the **Sample name** field).
3. Examine the **p-value** column for peak clusters that display statistically significant differences between groups.
4. Assess the quality of the underlying peaks using the **Plot Spectra in Cluster** feature. Manually inspect peak clusters with good P values for peak quality (good signal-to-noise ratio, resolution, accurate peak labeling, etc.) using the **Spectrum Viewer** and the 3 scatter plots.

Measure Biomarker Sensitivity and Specificity Using ROC Curves

1. Use the **P-Value Wizard** to automatically construct ROC plots and calculate AUC values.
2. Examine the **ROC area** column of the cluster table to identify peak clusters with good AUC values.

Assessment of Sample Relationships

Examine Sample Hierarchical Clustering Patterns

1. Select the sample group for examination using the **P-Value Wizard**.
2. Launch the **Heat Map** feature to visualize peak clusters as green and red points with variable shades based on relative expression level.
3. Use the dendrogram to create a clustered heat map that illustrates sample relationships.
4. Examine the clustered heat map for correlations between patterns of samples and the grouping criterion.

Visualize Sample Similarity Using PCA

1. Select a grouping criterion using the **P-Value Wizard**.
2. Launch the **PCA** feature to automatically generate PCA plots.
3. Examine the distribution of the samples in the PCA plots to identify possible associations with the grouping criterion.



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